

The Physical State and Its Geometry:

a certified, gauge-free formulation of quantum matter,
from declared content to the continuum problem

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Abstract

A quantum theory of gauged matter is a single object, a state ω_0 : the minimizer of a linear energy functional over a unital $*$ -algebra of observables $\mathcal{A}$, an exact quotient with no gauge sector. Every quantity the theory emits is a certified two-sided bracket on a conic linear program over the state's moment cone: a finite sum-of-squares certificate below, an exhibited admissible state above, sound at every stated budget independently of the convergence of any numerical process. The width of every emission is itself a read of the state, its distance in the certificate's own seminorm from an exposed face of the cone, and Heisenberg uncertainty and the exclusion principle are the two graded instances of one positivity relation.

From the state we construct, canonically and without choices, its carrier, closure ideal, modular structure, Dirichlet form and Dirac operator, metric, and spectral dimension and volume: time, scale, metric geometry, order, and frame are outputs of the state, not inputs to the theory. A field–gauge correspondence identifies compact-gauge and $\text{CAR} \otimes \text{CCR}$ content as two generating views of one dual-pair carrier, the Gauss law their singlet compression; on it the group's rank is an output of the closure ideal and, conditionally, the signature of spacetime is a modular output. Space is proposed as the modular structure of a committed net built from the state's own metric, with four decisive tests specified in advance and none yet executed.

The thermodynamic limit is proved for energy densities, with a finite-size gap criterion available on the stoquastic class, whose boundary is the fermion sign problem, stated here as an open structural question. The continuum limit is reduced to a certification problem: refinement acts on moment bodies by exact linear projection, failure of certified contraction must localize at a genuine zero-slack state, no accumulation of certified data can decide between the branches, and, conditionally on a single parametric certificate at the Gaussian fixed point and a finite chain of step certificates, the continuum brackets of an asymptotically free theory constitute a finite, machine-verifiable problem. The existence of the chain is the central conjecture: nonperturbative asymptotic freedom in certificate form.

A fully worked instance, $\text{SU}(2)$ gauge theory on two plaquettes, supplies certified spectra, closed-form geometry, and coupling towers, and passes the framework's first decisive test: its Weyl consistency requirement, certified spectral volume against the Hausdorff volume of the state's metric, closes exactly, both sides in closed form. Electrodynamical matter, with gravitation carried as a spin-2 boson sector, is developed as declared content of the same framework; its reads recover spectroscopy, order, frame, response, and the measured gravitational quantities, and an executed relational control in hydrogen exercises the emission pipeline end to end on a case where both numbers are independently known.

1 Introduction

Three commitments define this formulation.

First, the state is the only unknown. A theory is declared by its content: an algebra of observables presented as an exact quotient (the gauge sector removed at the level of the algebra, never fixed), a canonical family of derivations induced by the content’s symmetry, and a Hamiltonian element. Everything else (the carrier Hilbert space, the ground representative, the flow of time, the metric, the dimension, the organization of matter) is constructed from the state or read from it. Where the construction requires an unproven statement, that statement appears below as a named hypothesis, conjecture, or open problem, in place, with the results that depend on it stated conditionally.

Second, every emission is a certified two-sided bracket. The instrument never emits an estimate. A lower bound is a sum-of-squares certificate over the algebra, verified as a finite exact identity; an upper bound is the exact evaluation of an exhibited admissible state. Both are valid at every stated budget regardless of how they were found; no optimizer’s convergence is load-bearing for truth. Emitted quantities are dimensionless and relational; dimensionful intermediates never leave the instrument.

Third, nonlinearity in the state is the signature of a gauge mode. Every physical functional in this paper is linear in the state, an element of a convex cone, or an envelope of affine functions of the state (the supporting-hyperplane structure of the moment cone made visible); wherever a construction appeared to require a nonlinear object (a wavefunction, a chart, a threshold, a rescaled block field), the nonlinearity marked a representation artifact, and the construction is given here in its linear form.

Vocabulary

The paper uses a small fixed working vocabulary throughout; its nearest standard equivalents are these.

term	nearest standard reading
content	a presented theory: algebra (exact quotient), canonical derivations, Hamiltonian element, budget axes
budget	a declared finite truncation axis (word degree, spectral cutoff, band, volume)
read	a derived quantity, linear (or an affine envelope) in the state
emission	a certified two-sided enclosure $[L, U]$ of a read
bracket	the interval $[L, U]$ itself
certificate	a finite sum-of-squares identity with multipliers, verified in exact or interval arithmetic
width	$U - L$, itself a read of the state (Theorem 2.10)
instrument	the certification pipeline as a whole
fails shut	returns no bound rather than an unsound one
rung	a level of the constructive trial ladder (§2.3)
pin	an executed check against an independently known value

Reading order

Sections 2–3 are the framework and are self-contained; a reader who wants the framework exercised can pass from §2 directly to the worked instance of §5 and its appendices. §§8–9 carry the complete-proof analysis of the thermodynamic and continuum questions; §6 is classical invariant theory and Howe duality read into the frame; §7, with the real-form discussion it carries, is the conditional register of the paper, committed constructions with named decisive tests and nothing exhibited; §11 is the framework evaluated at electrodynamic content and can be read independently after §2. Proof statuses are graded in §12.

The paper proves what can be proved and states precisely what cannot yet be. The main results are: the uncertainty hierarchy that identifies every emitted width with the state’s distance, in the certificate’s own seminorm, from an exposed face of the moment cone, and places Heisenberg uncertainty and the exclusion principle as the two graded instances of one relation (§2.5); the constructive upper bracket, a single trial ladder whose reference rung is the kinetic vacuum read at the content’s group: Haar for gauge content, quasi-free⊗Gaussian for field content, one construction at two groups (§2.3); the field–gauge correspondence, on which compact-gauge and field content are two generating views of one dual-pair carrier, the Gauss law is singlet compression, and the group’s rank is an output of the state’s closure ideal, the spacetime group entering as its complex point with the real form a modular output (§§6, 7); the canonical construction stack of §3, including the automatic local factorization of stoquastic-class Hamiltonians in the ground carrier (Theorem 3.4), with the fermion sign problem identified as its exact scope boundary (Open Problem 3.6); the geometric identification and its falsifiable Weyl consistency requirement, verified in closed form on the worked instance (§4, Theorem 4.2); complete extensivity theorems for energy densities and a conditional finite-size gap criterion (§8); the reduction of the continuum problem to a finite certification problem, comprising the exactness of refinement (Proposition 9.1), the two-dimensional theorem (Theorem 9.5), the dichotomy and localization theorems (Theorems 9.9, 9.13), the no-go theorem on data (Theorem 9.14), and the conditional Reduction Theorem (Theorem 9.10), with the central conjecture (Conjecture 9.16) stated in minimal form (§9); the hadronic content and its certified targets (§10); and electrodynamic matter developed as declared content, with its single-emission structure, convergence theorems, gravitational reads, and executed hydrogen control (§11). A worked instance with certified numbers runs through the paper (§5).

2 Kinematics and emissions

2.1 Content

A theory is declared by: (i) a unital $\ast$ -algebra $\mathcal{A}$, presented as an exact quotient so that no gauge degree of freedom exists in $\mathcal{A}$; for lattice gauge content with compact group G on a declared face set, $\mathcal{A}$ is the algebra of invariants (generated by loop words), and the quotient requires no gauge fixing and no section; (ii) the canonical derivations $\{X_a\}$: the action of the Lie algebra $\mathfrak{g}$ on $\mathcal{A}$ induced by the content’s group action, with kinetic form the Casimir of the action (unique up to the normalization quotiented below); (iii) a coupling λ and a Hermitian magnetic element $M \in \mathcal{A}$, assembling the Hamiltonian $h(\lambda) = \text{Cas} - \lambda M$; (iv) budget axes: the canonical filtrations of $\mathcal{A}$ by degree over the canonical generators and by spectral cutoff of the Casimir. Nothing else is input. In particular no Hilbert space, metric, dimension, lattice spacing, coordinate, or basis is declared. The group in (i) and (ii) enters with the standing of

a presentation, not of an independent input: on the correspondence of §6 the same content is declared by its fusion functional alone, the pairing symmetry of the words carrying the content and the rank recovered from the closure ideal of the state (Proposition 6.5); the group is a choice of generators, and emissions do not depend on it (exactly at the worked content, Proposition 6.3; per Open Problem 6.4 in general).

2.2 States, cones, certificates

States are normalized positive linear functionals on $\mathcal{A}$. At budget b a state restricts to its degree- $\leq b$ moment data. Two objects at budget b must be held apart. The *moment body* $\mathcal{S}_b$ is the set of such restrictions over admissible states. The *moment cone* $\mathcal{M}_b$ is its computable outer approximation: the convex set of degree- $\leq b$ moment data satisfying the linear matrix inequalities expressing positivity on squares. The inclusion $\mathcal{S}_b \subseteq \mathcal{M}_b$ is generally strict at finite b ; its defect is exactly what Theorems 2.10 and 2.11 quantify, and conflating the two objects would collapse the hierarchy. Certificates are elements of the dual cone: finite sum-of-squares identities with multipliers over the declared constraint set. An *emission* is a certified interval $[L, U] \ni v$ for a value

$$v = \inf_{\omega \in \mathcal{S}} \omega(A) \quad (\text{or sup}),$$

a linear functional optimized over a convex set $\mathcal{S}$ of admissible states: L from a certificate by weak duality, U from an exhibited member of $\mathcal{S}$ evaluated exactly.

Proposition 2.1 (Weak duality; anytime validity). *If $A - \lambda \mathbf{1} - \sum_i \mu_i C_i = \sum_j S_j^\dagger S_j$ holds as a finite identity in $\mathcal{A}$, with C_i the constraint elements vanishing in expectation on $\mathcal{S}$, then $\omega(A) \geq \lambda$ for every $\omega \in \mathcal{S}$. If the identity holds up to a residual R of bounded norm, then $\omega(A) \geq \lambda - \|R\|$. Both statements hold for arbitrary coefficient tuples; consequently every solver iterate emits a valid bound, and no convergence claim is load-bearing.*

Proof. Apply ω : squares are nonnegative by positivity, constraints vanish by admissibility, and $|\omega(R)| \leq \|R\|$ by the C^* norm bound. $\square$

Remark 2.2 (Relation to the moment–SOS and bootstrap literatures). The lower side of every emission is a noncommutative moment–sum-of-squares bound, and the architecture (positivity of moment matrices below, exhibited states above) is shared with active programs: the variational reduced-density-matrix method [2, 3], the Navascués–Pironio–Acín hierarchy [4], and the Hamiltonian and lattice bootstrap [5, 6, 7, 8]. That line has advanced well past its first papers: Euclidean lattice Yang–Mills has been bootstrapped at finite N , including $SU(2)$ in two, three, and four dimensions with two-sided margins on the plaquette free energy at the 0.1% level against Monte Carlo [9], with $SU(3)$ and abelian content following [10, 11]; matrix quantum mechanics has reached high-precision ground-sector correlators including D0-brane/BFSS content [12, 13], whose gravitational content is holographic (the bounded matrix theory is dual to a black hole); gravitation enters the present paper in an unrelated and more modest sense, as declared linear spin-2 content with a stated exposure (§11.7, Remark 11.17), and no holographic claim is made anywhere; and the quantum-information side has produced ground-energy lower bounds and equilibrium algorithms with certified convergence guarantees [14, 15]. The present formulation is a member of this family, and its worked instance (§5) is deliberately the smallest nonsolvable case, not a competitive computation. What it adds is: the Hamiltonian (Kogut–Susskind) formulation posed directly on the gauge-invariant coordinate ring, the region

constraint being the second fundamental theorem in person (§6), where the cited lattice-gauge results are Euclidean; the certification discipline, finite exact-rational identities in the lineage of [16] with anytime validity (Proposition 2.1), where the cited programs report the output of floating-point conic solvers; the two-sided rule that every emission carry an exhibited admissible state on its upper side; the relational emission doctrine; and the superstructure of §§3–9 (geometry, extensivity, refinement) erected on the same cone. The finite- N Euclidean $SU(2)$ margins of [9] are the natural cross-community contact point for the Hamiltonian brackets emitted here, once the two formulations are related (anisotropic-limit matching).

Definition 2.3 (Boson word calculus). When the content carries bosonic (Weyl/CCR) sectors the unbounded canonical operators forbid a naive sum-of-squares reading, and certificate identities are instead required to live in the $*$ -algebra of boson-polynomial words under the syntactic constraint that every term of positive boson degree appear inside an exactly matched *designated square* of boson degree at most two per sector: a number form $b^\dagger b$ for any symplectic (Bogoliubov) image $b = ua + va^\dagger$ of the budgeted family (the original modes at $v = 0$); a completed square

$$-(Ba^\dagger + B^\dagger a) = \varepsilon\omega a^\dagger a + (\varepsilon\omega)^{-1}B^\dagger B - (\sqrt{\varepsilon\omega}a + B/\sqrt{\varepsilon\omega})^\dagger(\sqrt{\varepsilon\omega}a + B/\sqrt{\varepsilon\omega})$$

with B bounded; or a product of designated forms across commuting sectors ($[a, b] = 0$ gives $(a^\dagger a)(b^\dagger b) = (ab)^\dagger(ab)$). The verifier checks the patterns exactly; residual words are boson-degree-zero, and the anytime bound of Proposition 2.1 applies to them alone. Admissibility carries the finite-moment proviso per sector, bounded through the restricted covariance’s spectrum.

The physical state is selected linearly: ω_0 minimizes $\omega \mapsto \omega(h)$ over admissible states. The *ground sector* is the set of admissible states saturating the certified ground energy; operationally it is the zero-temperature end of the completely passive states [1]. For a nondegenerate gapped ground state the sector is a point.

Hypothesis 2.4 (Regularity, per content). $h(\lambda)$ is essentially self-adjoint on the canonical core; the ground state is unique on the systems considered; the quadratic form $\mathcal{E}_\omega$ of §3.3 is closable. These are standard and are verified per instance; all results below assume them where used.

2.3 The upper bracket: the trial ladder

Weak duality supplies the lower side of every emission; the upper side is the exact evaluation of an exhibited admissible state, and it is as constructive as the lower. An exhibited state meets four requirements, the trace of the tower’s own first level: (T1) it is admissible and saturates the ground energy e_0 , or averages to such a state at unchanged value under the content’s compact symmetry; (T2) its per-covolume moments of local words are finite exact contractions with certified rounding (uncertified tails are forbidden, tails as such are not); (T3) one construction serves every content; (T4) no object outside the theory enters, the class being generated from a reference determined by the content by operations internal to $\mathcal{A}$. The reference is one object, read at the content’s group.

Proposition 2.5 (The reference rung is the kinetic vacuum). *The reference state is the ground state of the kinetic Hamiltonian $h(0) = \text{Cas}$: the invariant vacuum of the content’s symmetry, determined by the content alone with no choice, whose moment functional is that symmetry’s*

singlet-channel fusion: the value of ω_{ref} on a product of words is the trivial-representation content of the corresponding tensor product. For compact gauge content this is the Haar state, its moments computed by character fusion $\int_G \chi_{j_1} \cdots \chi_{j_n} dU = \dim \text{Inv}(V_{j_1} \otimes \cdots \otimes V_{j_n})$ (§5); for CAR $\otimes$ CCR field content it is the quasi-free $\otimes$ Gaussian state, its moments computed by Wick contraction (§11). These are one construction at two groups: the pair contraction is the Heisenberg (or Clifford) group's singlet projector on $V \otimes V$ exactly as character orthogonality is the compact group's, so Wick's theorem and character fusion are the same operation: summing the singlet channels of a tensor product.

Remark 2.6 (The compact seam). For a compact group the invariant vacuum is the tracial Haar state; for a noncompact field sector it is the Fock vacuum of the distinguished kinetic representation. The difference is in what the vacuum is (a trace versus a lowest vector), not in the construction: both are the $h(0)$ ground state and both carry the singlet-fusion moment theorem. This is the sole place the one construction touches compactness; it is not a second construction. §6 gives the seam structural form: compact and field content are the two members of a reductive dual pair on one carrier, and the compression between them is the Gauss law.

Definition 2.7 (Enrichment class and trial ladder). The enrichment class $\mathfrak{E}$ is the automorphisms of $\mathcal{A}$ commuting with the content's symmetry and *strictly local* (each local element maps to a local element within a finite radius), graded by radius; concretely they are generated by the strictly-local elements from which the magnetic element M is built: plaquette (Wilson-loop) operators for gauge content, finite-radius charge-conserving circuits for field content, one operation with M the only input. Rung r of the trial ladder is the orbit of the reference rung under radius- $\leq r$ elements of $\mathfrak{E}$, conditioned by the character integrals of any sharp sectors; $U^*(r)$ is the ground-sector infimum of the read over rung r , non-invariant members averaged. Moments are exactly evaluable at every rung: conjugation of a local word through a radius- r element is exactly finite (a Pfaffian, a character fusion, or a Gaussian characteristic function, closed for symplectic and mixed-even gates, norm-convergent with absorbed tails beyond) at cost $e^{O(r)}$ poly in the lightcone volume.

Proposition 2.8 (Determination). *The trial ladder is the unique class satisfying (T1)–(T4) with intrinsic richness grading: the reference is forced by Proposition 2.5, the operations by the trace of (T1)–(T4), the grading is the enrichment radius. Each familiar alternative fails a named requirement. A similarity-transformed expectation that is not a state evaluation emits no bound (T1); a non-truncating unitary transform is not exactly evaluable (T2); a stochastic estimator carries error bars, not certificates (T2); a tensor network exactly contractible only in one dimension fails (T3), and elsewhere its approximate contraction bounds nothing (T2); a density-functional model is a fitted surrogate emitting no bound (T1, T2). The licensing runs forward: any exhibited admissible ground-sector state with exactly evaluable moments emits a valid bound, on or off the ladder.*

2.4 Residual nonlinearity, classified

Exactly two kinds of nonlinearity in the state survive audit, and both are linear structure in disguise: (i) *envelopes*: $E_1(\lambda) = \inf_{\omega} [\omega(\text{Cas}) - \lambda \omega(M)]$ is a lower envelope of affine functions of ω ; its concavity in λ and the Hellmann–Feynman relation $dE_1/d\lambda = -\langle M \rangle$ are the supporting-hyperplane structure of the moment cone; (ii) *content algebra*: polynomial relations among canonical generators (e.g. the cubic relation of §5) are algebraic geometry of the content, not

of the state. Anything else nonlinear (wavefunction representatives, field thresholds, rescaled block variables) is a chart and is excluded from definitions and emissions.

2.5 The uncertainty hierarchy

Membership in the moment cone is a hierarchy of uncertainty relations, and the width of every emission is a read of the state's distance from its boundary. Fix the degree- $\leq n$ word space $\mathbb{W}_n = \text{span}\{w_\alpha\}$ ($\dagger$ -closed, $w_0 = \mathbf{1}$) and the level- n moment matrix $M_n(\omega)_{\alpha\beta} = \omega(w_\alpha^\dagger w_\beta)$.

Theorem 2.9 (Generalized uncertainty at level n). *Write the centered Gram $G_{\alpha\beta}^c = \omega(\delta w_\alpha^\dagger \delta w_\beta)$, $\delta w_\alpha = w_\alpha - \omega(w_\alpha)$, and split it into Hermitian symmetric and commutator parts $G^c = \Sigma + \frac{1}{2}K$ with $\Sigma_{\alpha\beta} = \frac{1}{2}\omega(\{\delta w_\alpha^\dagger, \delta w_\beta\})$, $K_{\alpha\beta} = \omega([\delta w_\alpha^\dagger, \delta w_\beta])$. Then level- n admissibility $M_n \succeq 0$ is exactly $\Sigma \succeq -\frac{1}{2}K$, and straightening the canonical relations evaluates K as an affine function of moments of degree at most $2n - 2$. At $n = 1$ on a bosonic sector K is the central symplectic form and this is the Robertson–Schrödinger relation; at $n = 1$ on a fermionic sector the anticommutator matrix is central and $M_1 \succeq 0$ is the Coleman constraint $0 \preceq \gamma \preceq \mathbf{1}$. Heisenberg uncertainty and the exclusion principle are the two graded instances of one statement, and level n is its degree- n extension.*

Proof. $w_\alpha^\dagger w_\beta = \frac{1}{2}\{w_\alpha^\dagger, w_\beta\} + \frac{1}{2}[w_\alpha^\dagger, w_\beta]$ gives $G^c = \Sigma + \frac{1}{2}K$ after centering (a Schur complement in the unit entry); Σ and K are Hermitian, so $G^c \succeq 0 \iff \Sigma \succeq -\frac{1}{2}K$, identified with $M_n \succeq 0$ by centering. The commutator of degree- d_α, d_β words straightens to degree $\leq d_\alpha + d_\beta - 2 \leq 2n - 2$. $\square$

Theorem 2.10 (Gap identity). *For a level- n certificate with Gram $\Lambda = \sum_i c_i f_i f_i^\dagger \succeq 0$ (factors $F_i = \sum_\alpha (f_i)_\alpha w_\alpha \in \mathbb{W}_n$) and emitted value λ , and any admissible ground-sector state ω_0 with carrier vector Ω_0 ,*

$$\omega_0(X) - \lambda = \sum_i c_i \|\pi(F_i)\Omega_0\|^2 = \langle M_n(\omega_0), \Lambda \rangle_F.$$

The shortfall of any certificate on the physical state is the Λ -weighted squared residual of the state under the certificate factors: the squared distance, in the certificate's own seminorm, of the state from the kernel variety $\{F_i\psi = 0\}$. The emitted width is this distance, not a generic numerical residue.

Proof. Evaluate the certificate identity $X - \lambda \mathbf{1} - (\text{data multipliers}) = \sum_i c_i F_i^\dagger F_i$ in ω_0 : the energy and data multipliers vanish on the admissible ground sector, and $\omega_0(F_i^\dagger F_i) = \|\pi(F_i)\Omega_0\|^2 = f_i^\dagger M_n(\omega_0) f_i$ sums to the Frobenius pairing. $\square$

Theorem 2.11 (Closure is boundary contact). *The bracket closes at level n , $L_n^* = G^*$, iff the physical moment matrix sits on the face of the level- n cone exposed by the optimal certificate: for an attained optimum (λ^*, Λ^*) , the vanishing pairing $\langle M_n(\omega_0^*), \Lambda^* \rangle_F = 0$ with both factors positive semidefinite forces $M_n(\omega_0^*)\Lambda^* = 0$, so every certificate factor annihilates the state. This generalizes minimum uncertainty (degree-one annihilation) to a degree- n hierarchy: the degree-one kernel states are the squeezed and coherent vacua; the degree-two boundary already contains the Fock states $((a^\dagger a - m)|m\rangle = 0)$, all squeezed states, and the even/odd cat states ($a^2 - \alpha^2$ annihilates the even cat), on which level-two brackets close exactly. A ground state annihilated by no degree- $\leq n$ system compatible with the multipliers has positive width at every finite certificate, and the tower's convergence is the statement that these residuals fall to zero as n grows.*

Proof. $\text{Tr}(AB) = 0$ for $A, B \succeq 0$ gives $AB = 0$; applied to Theorem 2.10 at the extremal state, $f_i^\dagger M_n f_i = 0$ for each $c_i > 0$, i.e. $\|\pi(F_i)\Omega_0^*\| = 0$. The converse is the gap identity read on an annihilating certificate; without attainment the statement holds in the supremum, and every finite certificate still emits a valid bound. $\square$

Remark 2.12 (The width is not a local defect). A local uncertainty defect is not the emitted width. A two-mode Gaussian ground state whose global normal-mode annihilators are the certificate factors has certified width 4.9×10^{-6} at its stated budget (the bracket closes as the budget grows), while the Robertson–Schrödinger defect of its reduced single mode is $\nu^2 - \frac{1}{4} = 0.1406$: Theorem 2.10 governs, and its distance is measured in the certificate seminorm, whose active directions the multipliers select, never in a fixed local metric.

The annihilating factors of Theorem 2.11 are exactly the degree- $\leq n$ elements of the Gelfand ideal N_ω of §3: exactness is ideal membership. In the Hamiltonian-bootstrap literature the same phenomenon appears as the null-state condition [22]. The quantitative converse (a lower bound on the width by a fixed-metric distance to the exposed faces, with constants uniform over reads and budgets) is an error-bound question for the noncommutative hierarchy, and is not claimed.

2.6 Resolving emissions

A bracket that contains a target value confirms nothing by itself. An emission *resolves* a comparison only when its certified width is below a threshold declared in advance of the computation; thresholds may be tightened before a computation and never loosened after one. All comparisons in this paper are subject to this rule.

2.7 Execution and cost

The instrument is a constructor, not an existence argument: every emission is produced, and its execution has exactly four kinds of step, none of them a self-consistency loop. (1) One direct-optimization stream per certificate pullback, a search on a merit function whose convergence is not load-bearing (Proposition 2.1). (2) Machine-precision primitives that terminate: Gauss–Hermite integrals, characteristic functions and Pfaffians, character fusions, linear solves. (3) A finite verification pass in exact or interval arithmetic, matched patterns (Definition 2.3) checked syntactically. (4) Monotone budget sweeps with warm starts. The certificate multiplier is carried as a Gram $\Lambda = FF^\dagger$ with the factors transient; the scalar multipliers restore by closed-form least squares given F ; the climb maximizes the verified emitted value, unconstrained over F , with rank a monotone budget axis. That no step is a fixed-point iteration is what makes soundness independent of convergence: a valid two-sided bracket is in hand at every step of every stream, and the streams only narrow it.

Theorem 2.13 (Per-sweep optimality). *With M the generator-family size per covolume, the level- n certificate identity for a fixed matrix element has $\Theta(M^{2n}/(2n)!)$ independent coefficients up to the content’s word-symmetry factor (a constant of the content), each verified once, and the climb’s $O(M^{2n})$ per sweep is input-linear in that count, optimal up to rank and constants. A conditional projection onto a commuting subalgebra acts within the fixed word family and does not change the count; the energy-saturation multiplier adds one scalar. The counting is content-blind: each specialization of a content preserves the scaling class of the one it refines, paying an additive family count. Total time is $I(\epsilon) \cdot O(M^{2n})$, with $I(\epsilon)$ the number of sweeps spent to reach emitted width ϵ , a measured quantity carrying no a priori bound, so no unconditional total-time*

claim is made; natively sum-of-squares pieces (reciprocal kernels, matched boson squares) verify below the generic floor.

3 Canonical structures of the state

Throughout, ω_0 is the ground state of declared content under Hypothesis 2.4.

Remark 3.1 (Convention on canonicity). “Canonical” below always means: determined by the content and the state up to an equivalence that is named at the construction and that acts trivially on every emission. Emissions are values of the state on algebra elements; unitary equivalence of the carrier and the inner ambiguity of modular data leave every such value fixed, so the residual equivalence classes are gauge that never reaches an emission. Wherever a construction below depends on more than (content, ω_0) modulo such an equivalence, that dependence is stated, not absorbed into the word.

3.1 Carrier and closure

The GNS construction gives the carrier $(\pi, \mathcal{H}_\omega, \Omega)$, unique up to unitary equivalence. The *Gelfand ideal* $N_\omega = \{a \in \mathcal{A} : \omega(a^\dagger a) = 0\}$ is determined by the state alone. By §2.5, N_ω is the closure structure of the instrument: brackets close exactly on certificates whose factors lie in it.

Hypothesis 3.2 (Faithfulness). ω_0 is faithful on the abelian (multiplication) subalgebra. This is the chart-free residue of the classical statement “the ground wavefunction is strictly positive”; it is verified on the instances considered.

3.2 Modular structure

For a subalgebra $\mathcal{N} \subseteq \pi(\mathcal{A})''$ on which Ω is cyclic and separating, Tomita–Takesaki theory yields the modular data $(\Delta_\mathcal{N}, J_\mathcal{N})$ and flow, canonical up to inner automorphisms [35]. A pure ground state is separating for no global type-I algebra; modular structure therefore lives on *subalgebras*, and the admissibility criterion is internal: $\mathcal{N}$ is admissible precisely when $\mathcal{N} \cap N_\omega = \{0\}$. Time, in this formulation, is the modular flow of admissible subalgebra–state pairs; it is an output.

3.3 Dirichlet form, Dirac operator, factorization

The canonical derivation is $\partial f = (Xf)\Omega \in \mathcal{H}_\omega \otimes \mathfrak{g}$, valued in the Clifford module of $(\mathfrak{g}, \text{Killing})$; the *Dirichlet form of the state* is

$$\mathcal{E}_\omega(f, g) = \sum_a \omega((X_a f)^*(X_a g)),$$

manifestly linear in ω . Its generator structure is the Cipriani–Sauvageot theory [36]; the choice of first-order differential structure on which that theory depends is not made downstream: the derivations are declared content (§2.1), so the construction is functorial in content, and its content dependence is the dependence every object of the theory has. The associated Dirac operator D_ω satisfies $D_\omega \Omega = 0$ by construction.

Theorem 3.3 (Harmonicity). *Let the content be of Dirichlet type (the class of Theorem 3.4 below) and assume Hypotheses 2.4–3.2. Let ω be an admissible state whose dynamics is implemented on its carrier: a self-adjoint H_ω with Ω in its form domain and $e^{itH_\omega}\pi(a)e^{-itH_\omega} =$*

$\pi(\alpha_t(a))$ on the canonical core. Then the identity $\mathcal{E}_\omega(f, g) = \langle f\Omega, (H_\omega - E_0)g\Omega \rangle$ holds on the core if and only if ω is the ground state. The diagonal defect at $f = g = \mathbf{1}$ is $\omega(h) - E_0$: a nonnegative affine read of the state (the geometricity defect at unit), vanishing exactly on the ground sector.

Proof. If ω is the ground state, the identity is Theorem 3.4. Conversely, evaluate the claimed identity at $f = g = \mathbf{1}$: the left side vanishes because $X_a\mathbf{1} = 0$, and the right side is $\langle \Omega, (H_\omega - E_0)\Omega \rangle = \omega(h) - E_0$, so $\omega(h) = E_0$ and ω lies in the ground sector; uniqueness (Hypothesis 2.4) identifies it as the ground state. $\square$

Theorem 3.4 (Automatic local factorization, stoquastic class). *Let $h = \text{Cas} - \lambda M$ be of Dirichlet type with respect to the declared derivations: in the content's function realization, h is an elliptic Schrödinger operator (Casimir Laplacians per link plus a real multiplication potential), so that the ground representative ψ_0 is strictly positive (Perron–Frobenius; this is Hypothesis 3.2 holding by theorem rather than by assumption, and pure gauge content is in the class). Then, in the ground carrier,*

$$\langle f\Omega, (h - E_0)g\Omega \rangle = \sum_a \omega((X_a f)^*(X_a g)), \quad \text{i.e.} \quad h - E_0 = \sum_a L_a^\dagger L_a,$$

with $L_a(f\Omega) = (X_a f)\Omega$ local to link a and $L_a\Omega = 0$ for every a .

Proof. On the class stated, e^{-th} is positivity improving and $\psi_0 > 0$ (Perron–Frobenius for Schrödinger operators on products of compact group manifolds). The ground-state (Doob) transformation, standard in Dirichlet-form theory [21], then gives $(J_a - i(X_a\phi))(g\psi_0) = -i(X_ag)\psi_0$ for $\phi = -\log \psi_0$, by the Leibniz rule; summing adjoint squares yields the displayed identity, whose two sides are chart-free and linear in ω , hence hold in the carrier independently of the realization. Locality holds because X_a acts on the tensor factor of link a ; $L_a\Omega = 0$ because $X_a\mathbf{1} = 0$. Domains: the identity holds on the canonical core, where both sides are densely defined closable quadratic forms; Hypothesis 2.4 (essential self-adjointness and closability) identifies their closures, extending the identity to the form domains. $\square$

Remark 3.5 (Scope of the automatic precondition). On the stoquastic class of Theorem 3.4, the ground carrier automatically presents the Hamiltonian as a sum of local positive operators each annihilating the ground vector: the structural precondition of the finite-size gap criteria of §8, which generic Hamiltonians fail. The scope boundary is exactly the sign problem. Fermionic hopping content is not of Dirichlet type with respect to the declared derivations; the ground representative of such content is not positive in any local realization in general, and the obstruction cannot be removed by local basis changes; curing non-stoquasticity is computationally hard even when possible [23, 24]. For that content the transformation above is unavailable, and the identity between the ground form and the content Dirichlet form is not expected to hold.

Open Problem 3.6 (Local factorization for fermionic content). Determine for which fermionic contents the ground form admits any local positive decomposition $h - E_0 = \sum_x Q_x$ with $\Omega \in \ker Q_x$ for every x . This is the sign problem in carrier form. It is not an item completed by additional work of a known kind: a structural obstruction is possible, and either an existence argument or an obstruction theorem would be a result. Absent a decomposition, the certified finite-size gap criterion of §8 does not apply to fermionic content at moderate coupling, and the gap program for such content routes through the strong-coupling regime alone (§8.2). A candidate structural criterion is Conjecture 6.8 (§6).

3.4 Metric, dimension, volume, sectors

The carré du champ $\Gamma(f, g) = \frac{1}{2}[f \mathfrak{L}g + g \mathfrak{L}f - \mathfrak{L}(fg)]$, with $\mathfrak{L}$ the Casimir Laplacian, is content-canonical. Points are pure states of the abelian sector; the distance

$$d(p, q) = \sup\{p(f) - q(f) : \Gamma(f, f) \leq 1\}$$

is a conic linear program in the instrument's own certificate cone [37]. The counting function $N(E)$ of $D_\omega^2 = h - E_0$ is a linear read of spectral projections; spectral dimension is a certified window slope of N ; spectral volume is the Weyl coefficient, admissible as an emission only where Dixmier measurability (existence of the limit) is itself certified. Quantum-number sectors are the gradings under the automorphism group of the content; all emissions are invariant under them. On instances with unique ground state, $\dim \ker D_\omega = 1$.

4 The geometric identification

The identification made by this formulation is: *the certificate structure of the state, at closure, is a Dirac geometry*. The data $(\mathcal{A}, \mathcal{H}_\omega, D_\omega)$ constitute a spectral triple only if the axioms hold: compact resolvent of D_ω , bounded commutators $[D_\omega, f]$ on a dense subalgebra, regularity, and finite dimension spectrum [37]; a manifold requires in addition the reconstruction axioms [38]. None of these is assumed here. On finite content they are finite-dimensional trivialities (bounded commutators are the finiteness of $\Gamma(f, f)$ already used by the metric), and the substantive questions are their uniformity along extensivity (§8) and refinement (§9); each axiom is a named verification obligation per instance, on the same footing as Dixmier measurability in §3.4. The identification is falsifiable at its first quantitative joint:

Definition 4.1 (Weyl consistency requirement). For every instance, the certified spectral volume (the Weyl coefficient of $N(E)$, where measurable) must equal the Hausdorff volume of the spectrum under the metric d . Failure of this equality at certified resolution refutes the identification for the instance.

For the worked instance of §5 both sides are in closed form. The metric side (all values derived in Appendix A.1): the required volume is $(32 - 16\sqrt{2})\pi^2$, the diameter realization $d(\mathbf{1}, (-\mathbf{1}, -\mathbf{1})) = 2\sqrt{2}\pi$, and the one-loop distance $2|\arccos(x_1/2) - \arccos(x_2/2)|$. The spectral side is also closed form for this instance: every eigenvalue and multiplicity of D_ω^2 on the carrier is exact (Appendix A.4), so the eigenvalue count is exact arithmetic, the existence of the Weyl limit is proved rather than assumed (Dixmier measurability holds), and the comparison closes at every threshold of the resolution rule of §2.6. The comparison has been run, and it passes exactly:

Theorem 4.2 (Weyl consistency holds for the worked instance). *For the content of §5 the counting function of $D_\omega^2 = h - E_0$ satisfies*

$$N(E) = \frac{16 - 8\sqrt{2}}{3} E^{3/2} + O(E),$$

and the Weyl coefficient equals the Hausdorff volume of Appendix A.2 as an algebraic identity:

$$6\pi^2 \cdot \frac{16 - 8\sqrt{2}}{3} = (32 - 16\sqrt{2})\pi^2.$$

The spectral dimension read is 3, the dimension of the metric side. The leading coefficient is coupling-independent: $\|\lambda M\| \leq 6\lambda$ sandwiches N_{h-E_0} between shifted copies of the kinetic counting function, so the identity holds at every declared coupling. Proof in Appendix A.4.

The two closed forms share no machinery: the volume is a triple integral of the Γ -metric's volume element, the count is Clebsch–Gordan combinatorics; their agreement is exactly the equality Definition 4.1 was built to test, and it is an instance of the equivariant Weyl law [17, 18], verified directly. On a commutative carrier the identification *must* reduce to the classical Weyl law, so the content of the test is not the law but the normalizations: every constant entering either side was fixed elsewhere (the metric normalization by the verified carré du champ of §5, the operator by the declared content, the quotient measure by the submersion), with no adjustable element remaining at the comparison. A mismatch by any factor would have refuted the identification; there was none. No result elsewhere in the paper depends on this outcome; the geometric identification has now passed its first decisive test, and the continuum program of §9 stands or falls independently.

5 Worked instance: SU(2) on two plaquettes

5.1 Content and algebra

Gauge group SU(2) on two faces sharing a boundary; the invariant algebra is generated by the loop characters $x = \chi(U_1)$, $y = \chi(U_2)$, $z = \chi(U_1 U_2)$, freely up to the classical relation defining the character variety

$$\kappa(x, y, z) = x^2 + y^2 + z^2 - xyz - 4 \leq 0, \quad |x|, |y| \leq 2,$$

and the Hamiltonian is $h(\lambda) = C_2^{(1)} + C_2^{(2)} - \lambda(x + y + z)$. All moments of the Haar state are exact rationals computed by character fusion; the carré du champ closes on polynomials in (x, y, z) with rational structure constants (e.g. $C_2 x = \frac{3}{4}x$, $\Gamma(x, x) = (4 - x^2)/4$).

5.2 Certified spectra

Two-sided brackets on $E_1(\lambda)$, produced by sum-of-squares certificates with multipliers on the region above (lower side) and exact evaluation of exhibited states (upper side), verified in exact rational arithmetic:

λ	certified bracket on E_1	width
1/16, 1/12	two-sided, width $\leq 10^{-15}$	closed
1/2	$[-0.9548, -0.8836]$	0.071
2	$[-7.0500, -6.9595]$	0.090

The plaquette expectation obeys the certified bounds $4.0031 \leq \langle x + y + z \rangle_{\lambda=2} \leq 6$, the lower side from the concavity (supporting-hyperplane) structure of §2.4, the upper from the exact norm.

5.3 Coupling towers

Weak coupling (derived; Appendix A.6):

$$E_1(\lambda) = -\frac{10}{3}\lambda^2 - \frac{32}{9}\lambda^3 + \frac{551}{297}\lambda^4 + \frac{172064}{9801}\lambda^5 + O(\lambda^6),$$

with all coefficients exact and consistent with the certified brackets. *Strong coupling* (derived at leading orders; Appendix A.5): harmonic expansion about the identity class in the Γ -metric gives normal-mode stiffnesses $3\lambda/4$ and $\lambda/4$ (three components each), hence

$$E_1(\lambda) = -6\lambda + \frac{3}{2}(1+\sqrt{3})\sqrt{\lambda} + c_0 + O(\lambda^{-1/2}), \quad E_2 - E_1 = 2\sqrt{\lambda} + O(1),$$

gap-ratio asymptote $(E_3 - E_1)/(E_2 - E_1) \rightarrow (1 + \sqrt{3})/2$, plaquette law $\langle x+y+z \rangle = 6 - \frac{3}{4}(1+\sqrt{3})/\sqrt{\lambda} + O(1/\lambda)$ (consistent with Hellmann–Feynman by construction and with the certified bounds above); the invariant-sector selection rules place the swap-even excitation lowest at both ends of the coupling axis. The constant term is derived in Appendix A.5:

$$c_0 = -\frac{126 + 13\sqrt{3}}{192} \approx -0.77352,$$

inside the certified window -0.79 ± 0.05 at $\lambda = 2$ outright; fixing it determines subleading coefficients $(c_{-1}, c_{-2}) \approx (+0.07, -0.08)$ from converged variational values at $\lambda \in \{2, 8\}$, which then land the certified window -0.83 ± 0.04 at $\lambda = 1/2$ —one constant and two small corrections carrying four couplings.

6 The field–gauge correspondence: the group as output

Proposition 2.5 exhibited one construction at two groups. This section states the structure behind the seam of Remark 2.6 and draws its consequence: compact-gauge content and $\text{CAR} \otimes \text{CCR}$ field content are two generating views of a single carrier, related by a reductive dual pair; the Gauss law is the singlet compression between the views; and, on this correspondence, the symmetry itself moves from the input side to the output side. The group’s rank is recovered from the closure ideal of the reference state, the same object that closes brackets (Theorem 2.11), and the existence of a symmetry becomes a positivity question of the type the instrument already runs.

6.1 One carrier, two generating subalgebras

Definition 6.1 (Dual-pair carrier). Let $W = W_1 \oplus W_0$ be a $\mathbb{Z}_2$ -graded one-particle space and $\mathcal{F}$ the Fock space carrying CAR over W_1 and CCR over W_0 . The quadratic words close on the orthosymplectic algebra $\mathfrak{osp}(W)$, the kinetic symmetry of field content (the Bogoliubov transformations). A *reductive dual pair* $(\mathfrak{g}, \mathfrak{g}')$ is a pair of subalgebras of $\mathfrak{osp}(W)$, each the full commutant of the other [25]. The distinguished vector is the lowest-weight vector Ω of the oscillator representation, fixed by the compact stabilizer that the kinetic Hamiltonian’s spectral split determines; as in §11.4, the polarization is an output of the content, not a choice, and no fully $\mathfrak{osp}$ -invariant state exists to choose instead (for CCR none exists; for CAR it is the trace, not the Fock vacuum).

The *gauge view* generates from Ω by the $\mathfrak{g}$ -invariant words (loop words and characters); the *field view* generates by the linear field words. The kinetic term of the gauge view is the electric Casimir of $\mathfrak{g}$; that of the field view is the quadratic current form $\sum_a J_a J_a$ of the dual partner; the two Casimirs of a dual pair are related through the number operator by Capelli-type identities. Both views read one state.

Proposition 6.2 (Wick is fusion; the Gauss law is Haar integration). *The moments of Ω on $\mathfrak{g}$ -invariant words equal the singlet-channel fusion of the corresponding tensor products: both sides evaluate the projector onto the trivial isotypic component. The first fundamental theorem of invariant theory supplies the dictionary (pairings span the invariants of tensor powers: symmetric for orthogonal, antisymmetric for symplectic, mixed $V-\bar{V}$ for unitary content) [26], and the Weingarten calculus is its exact finite- N form: Haar integration of a word equals a signed sum of Wick pairings with explicit rational coefficients [27]. Consequently the compression $P = \int_G \pi(U) dU$ onto the singlet sector is Haar integration, is linear in the state, and forms no section: the gauge state is the field state compressed to invariant words, in the discipline of Remark 11.5. Provenance: classical invariant theory transported whole, numbered for the dependency graph (§12).*

6.2 Exactness and budget

Proposition 6.3 (Prepotential exactness for $SU(2)$). *Peter–Weyl gives $L^2(SU(2)) \cong \bigoplus_j V_j \otimes V_j$, the number-matched singlet sector of the Fock space of two oscillator doublets per link, one per link end [28, 29]. Under the identification the electric Casimir is polynomial in the number operator, $C_2 = (\hat{N}/2)(\hat{N}/2 + 1)$, and the loop characters are polynomial words in oscillator bilinears. The unitary link operator itself carries $(\hat{N} + 1)^{-1/2}$ normalizations and is rational, not polynomial; the instrument’s generating set is the polynomial intertwiner words, and the classical relation of §5 is carried by the closure ideal. The identification is exact: the worked instance of §5 is a singlet-compressed bosonic content with no truncation.*

For general compact G the known exact presentations are per representation: rishon and quantum-link content realizes the link algebra on a finite-dimensional fermionic carrier exactly at fixed representation [31], so the representation cutoff is a *declared budget*, and full $L^2(G)$ is its refinement limit along the machinery of §9, a new budget axis with the same exact projection structure (Proposition 9.1).

Open Problem 6.4 (The general dual-pair presentation). Construct, for each compact G of the declared contents, the dual-pair carrier and singlet compression presenting $L^2(G)$ -content exactly ($SU(2)$ is exact above; $SU(3)$ prepotentials with additional constraints are known [30]), and carry the canonical machinery of §3 through the compression. This construction routes Open Problem 7.4 and the gauge instance through one object: the Dirichlet form, derivations, and Γ -calculus of the gauge sector become Wick-computable, and the four tests of Hypothesis 7.3 and the gauge machinery stop being two machineries.

6.3 The group is an output of the state

Proposition 6.5 (Group from closure). *For the classical families ($U(N)$, $O(N)$, $Sp(N)$) the moment matrix of the reference functional on invariant words of degree $\leq b$ is the Gram matrix of the pairing (Brauer or partition-algebra) basis, with the rank N entering only as a parameter; the first fundamental theorem says these words span [26]. The family is fixed by the declared pairing symmetry of the words (orthogonal, symplectic, or unitary), and the rank is recovered from the Gelfand ideal of the functional: at integer N the ideal is generated by the relations of the second fundamental theorem, the determinantal and Pfaffian identities, entering at a degree that reads off N . Content may therefore declare the fusion functional, and the symmetry is read from the state by the instrument’s own closure test (§2.5) rather than declared. For the worked*

instance the recovered relation is the Fricke cubic of §5: the region constraint the instrument has been carrying is the second fundamental theorem in person, its inequality the real (positivity) locus.

Remark 6.6 (What the ideal cannot see). The output of Proposition 6.5 is the *complexification* and nothing finer. Invariant theory sees the rank and the symmetry type of the invariant form, never its signature: the Brauer data at $\delta = 4$ with symmetric pairing carry the same fundamental theorems, hence the same closure ideal, for $O(4)$, $O(3,1)$, and $O(2,2)$: the real forms of the one complex group $O(4, \mathbb{C})$ that preserve a real symmetric pairing. (The remaining real form, the quaternionic $O^*(4)$, preserves no real symmetric form and corresponds to no signature; it is addressed with the signature question in §7.1.) This is a scope statement, not a defect: real forms are classified by conjugacy classes of antilinear involutions, which no complex-linear moment datum can encode, and the theory’s canonical antilinear objects are the modular conjugations of §3.2. The resolution of the real form is therefore a modular question and is taken up in §7.1, in that section’s conditional register. Two finer points share this boundary: the ideal fixes O rather than SO , the orientation word (the ε -tensor, a Pfaffian invariant) not being in the pairing image, so orientation is a separate declared-or-read item; and the integer output is the parameter δ , identified with a dimension only through the read of §3.4.

Remark 6.7 (Quantization of symmetry as positivity). Away from the integer points the fusion functional on the pairing algebra is generically not positive: the interpolating categories exist and are semisimple at generic parameter but are not unitary [33], and in the Temperley–Lieb instance positivity holds exactly on the Jones set $\{4 \cos^2(\pi/k)\} \cup [4, \infty)$ [32]. The existence of a symmetry is thus a positivity question of the same linear-matrix-inequality type the instrument runs at every budget: *which* groups can exist is a moment-cone membership, and the integrality of N is an output. At generic large parameter the functional degenerates to the planar (free) functional, the master field of the large- N limit [34]; the certified interpolation between the integer points and the planar functional is unexplored territory of the same cone.

6.4 A structural candidate for the sign problem

Conjecture 6.8 (Dual-pair criterion). *A fermionic content admits the local positive ground decomposition of Open Problem 3.6 if and only if it is the singlet compression of a content in the stoquastic class of Theorem 3.4. The forward half (compressions of stoquastic content inherit a decomposition on invariant words) is the tractable half and should be settled first; the converse would make sign-problem freedom and hidden gauge structure one property. Either resolution sharpens Open Problem 3.6 from a case list toward a criterion.*

6.5 Decisive checks, specified in advance

None of the following has been executed. (a) *Prepotential re-pin*: re-derive the certified two-loop brackets of §5 in prepotential words; the E_1 windows and the metric normalization $\Gamma(x, x) = (4 - x^2)/4$ must reproduce to the digit, the same moment problem posed in different generators, and any discrepancy refutes the compression as implemented. (b) *Casimir dictionary*: verify $C_2 = (\hat{N}/2)(\hat{N}/2 + 1)$ as a finite-dimensional operator identity per truncation. (c) *Rank drop*: the certified kernel of the degree- $\leq b$ fusion Gram at parameter $N = 2$ must coincide with the degree- $\leq b$ slice of the second-fundamental-theorem ideal, the executable form of Proposition 6.5. (d) *Complex-point rank drop*: check (c) at symmetric pairing and parameter $\delta = 4$; the

certified kernel of the degree- $\leq b$ Brauer Gram must coincide with the degree- $\leq b$ slice of the rank-five determinantal ideal of $O(4, \mathbb{C})$, the executable half of Remark 6.6, with no Lorentzian input anywhere in the computation. Check (a) also bears on the strong-coupling constant of Appendix A.5: in prepotential variables the group-manifold measure is replaced by the flat Fock measure, so c_0 becomes a normal-ordering computation on polynomial operators—an independent route to the derived value and a cross-check on the compression itself.

7 Space from the state

Space is not declared; it must be constructed from $(\mathcal{A}, \omega_0)$. The known theorems run from nets of subalgebras to geometry: a half-sided modular inclusion generates a positive-energy translation group [39, 41]; families satisfying geometric modular action determine spacetimes and isometry groups [40]; modular flow of wedge algebras is geometric in the free field [42]. What is missing from the literature, and supplied here as the committed construction, is the map from $(\mathcal{A}, \omega_0)$ to the net.

Remark 7.1 (Motivation for the construction; not a uniqueness theorem). The imported theorems consume two structural features: a directed family of nested subalgebras admitting a common cyclic separating vector (half-sided inclusions), and a disjointness relation, i.e. pairs of members with commuting or state-independent generation at extensive content (the spacelike structure of geometric modular action). Among the structures the state defines in §3, the ball net of the spectral metric supplies both from outputs alone. The alternatives each lack at least one: the Gelfand ideal is a criterion, not an index; the modular data of the full algebra trivialize on a pure state; sector gradings are discrete, with no directed region structure; and the budget filtrations (including multi-parameter ones, which are genuinely partially ordered) are content-uniform, so distinct levels are nested and never disjoint, supplying direction without disjointness (partitioning the generators to manufacture disjointness re-declares regions, which is the input this section exists to output). This elimination is a survey of the structures the theory currently defines, not a theorem that no other index can exist; the construction below is therefore *committed* rather than derived, and the weight rests on the decisive tests of Hypothesis 7.3, not on this remark.

Definition 7.2 (The ball net). For metric balls B of (Spec, d) , let $\mathcal{A}(B)$ be the subalgebra of elements with derivation support in B . $\mathcal{A}(B)$ is admissible when $\mathcal{A}(B) \cap N_\omega = \{0\}$ (the state separating on it): the closure structure decides which regions carry geometry.

Hypothesis 7.3 (Geometric modular action of the ball net). The assignment $(\mathcal{A}, \omega_0) \mapsto \{\mathcal{A}(B)\}$ is functorial, and on extensive content its relative modular data satisfies geometric modular action, the emergent translations and isometries being those of the output geometry. *Scope*: on small bosonic instances the ball net localizes configuration space and no spatial claim is made; the spatial claim is made on extensive (many-body, fermionic) content.

Hypothesis 7.3 carries four decisive tests in the free chiral fermion, where every answer is known and no adjustable element exists: the metric of the vacuum's Dirichlet form must reproduce arc length; the ball net must reproduce the interval net; the modular flow of a half-space algebra must be the known dilation flow [42]; and the translations generated by nested half-space inclusions [39] must be the physical ones. Failure of any one refutes the hypothesis. None of the four has been executed: each requires the fermionic machinery of Open

Problem 7.4, which is not yet built; they are specified in advance so that failure is defined before the machinery exists. The division of labor is stated plainly: the imported theorems are standard, and the entire difficulty lives in their hypotheses (cyclic separating vectors, half-sided inclusions, geometric modular action); Hypothesis 7.3 is exactly the assertion that the ball net meets those hypotheses on extensive content. No instance of emergent geometry, not even a toy, is exhibited in this paper.

Open Problem 7.4. Construct the quasi-free (CAR) instance of the Dirichlet-form and derivation machinery of §3.3. This construction is prerequisite both to the tests above and to all fermionic content of §10. The correspondence of §6 routes this construction and the gauge machinery through one object (Open Problem 6.4).

7.1 The spacetime group: the real form as modular output

The correspondence of §6 outputs the spacetime symmetry only up to complexification: invariant words see the rank and pairing symmetry of a form, never its signature, so the closure ideal at parameter $\delta = 4$ fixes $O(4, \mathbb{C})$ and cannot distinguish its real forms (Remark 6.6). Real forms of a complex group are classified by conjugacy classes of antilinear involutions, and the canonical antilinear data of this theory is the modular conjugation $J_{\mathcal{N}}$ of admissible subalgebra–state pairs (§3.2). The division of the group is then exactly the division of the instrument: the compact real form is unitarized by the GNS carrier of the fusion functional, which is positive and tracial (the reference rung, the Euclidean side); the noncompact generators, the boosts that distinguish the Lorentz form, are carried by the modular flow of the ground sector, Δ^{it} of wedge-type members of the ball net, which is the content of the Bisognano–Wichmann theorem on the free field [42] and of geometric modular action in general [40]. On this reading the Lorentz group is never a declared symmetry: the closure ideal supplies the complex point, the modular conjugations select the involution, and Wick rotation is not a contour choice but the KMS analyticity of the modular read (§11). Signatures with two causal directions are excluded internally rather than by declaration: a half-sided modular inclusion generates positive energy along one causal direction [39], and no second independent inclusion of modular type is available, so $O(2, 2)$ -type forms fail the hypotheses. The quaternionic real form $O^*(4)$, which preserves no real symmetric form, is the homogeneous isometry group of no signature and therefore never arises as the output of geometric modular action on a metric ball net; it is excluded at the same step. The disconnected components (parity, time reversal) are likewise antiunitary, J -type data, not new inputs.

Definition 7.5 (Signature consistency requirement). On extensive content satisfying Hypothesis 7.3, the complexification of the homogeneous isometry group produced by the geometric modular action of the ball net must equal the complex group fixed by the closure ideal of §6 at parameter δ equal to the read spectral dimension. Every arrow in the chain (state to dimension read, dimension to δ , δ and ideal to complex point, modular conjugations to real form) is linear in the state or classical algebra; failure of the equality at certified resolution refutes the identification, exactly as Definition 4.1 refutes the geometric one.

The requirement is conditional on Hypothesis 7.3 and inherits its status: nothing here is exhibited, and the comparison is blocked on Open Problem 7.4 together with the four tests above, which now do double duty: the half-space modular flow test is simultaneously the test that the real form, the signature of spacetime, is an output of the state rather than an input to the theory.

8 Extensivity

8.1 Energy densities

Theorem 8.1 (Certified thermodynamic densities). *Let $H_\Lambda = \sum_{\ell \in \Lambda} C_2^{(\ell)} - \lambda \sum_{p \in \Lambda} \chi_p$ on boxes Λ (interior plaquettes), G compact, with $c_\chi := \|\chi_p\|$ (for the $SU(2)$ fundamental character $c_\chi = 2$; in general the dimension of the plaquette representation). Then $e(\lambda) = \lim E_0(\Lambda)/|\Lambda|$ exists, and for every Λ ,*

$$\left| e(\lambda) - E_0(\Lambda)/|\Lambda| \right| \leq \frac{c_\chi \lambda n_\partial(\Lambda)}{|\Lambda|},$$

with n_∂ the exact count of boundary-crossing plaquettes per tile. Certified finite-volume brackets on $E_0(\Lambda)$ therefore yield certified brackets on $e(\lambda)$ of width $(U - L)/|\Lambda| + 2c_\chi \lambda n_\partial/|\Lambda|$.

Proof. Tile $N\Lambda$ by N copies: electric terms partition exactly over links; magnetic terms split into interior and crossing, $H_{N\Lambda} = \sum_i H_\Lambda^{(i)} + W_N$, where W_N collects the crossing plaquettes, of which there are at most Nn_∂ (each crossing plaquette assigned to one of the tiles it meets), so $\|W_N\| \leq c_\chi \lambda N n_\partial$. Hence $E_0(N\Lambda) \geq N E_0(\Lambda) - c_\chi \lambda N n_\partial$ as an operator bound, and evaluating $H_{N\Lambda}$ in the product of copy ground states, $E_0(N\Lambda) \leq N E_0(\Lambda) + c_\chi \lambda N n_\partial$. For existence of the limit, apply the same interface estimate to the bipartition of a box into two sub-boxes: it shows $E_0(\Lambda) + c_\chi \lambda n_\partial(\Lambda)$ is subadditive under box concatenation along each axis, so Fekete's lemma gives convergence of $E_0(\Lambda_k)/|\Lambda_k|$ along side-doubling sequences; the two displayed operator bounds then sandwich an arbitrary box between the tilings it embeds in and those that embed in it, with per-volume error $c_\chi \lambda n_\partial/|\Lambda|$, which transfers the limit to arbitrary boxes and identifies it as box-shape independent. Dividing the two bounds by $N|\Lambda|$ and letting $N \rightarrow \infty$ gives both inequalities with the same constant. $\square$

8.2 Gaps

On the stoquastic class of Theorem 3.4, which includes pure gauge content, $h - E_0 = \sum_x Q_x$ with Q_x local, positive, and $\Omega \in \ker Q_x$ for all x : the ground carrier automatically presents the structure that finite-size gap criteria require [43, 44]. The quantitative content of those criteria is an overlap (angle) condition between local kernels, which is a moment read (Gram data) and hence certifiable. For fermionic content the precondition is not automatic and may be obstructed (Open Problem 3.6); the criterion of this paragraph does not reach such content. One transport step remains open on the stoquastic class:

Open Problem 8.2 (Carrier transport). The decomposition of Theorem 3.4 is taken in the carrier of the ground state of each volume. Establish certified bounds transporting the kernel angles of the decomposition on a larger volume from moment reads on a smaller one together with a certified overlap of the two ground states on local elements.

Given Open Problem 8.2, the martingale argument of [44] yields a certified finite-size gap criterion on the stoquastic class: certified block gaps and certified kernel angles imply an explicit infinite-volume gap bound. The strong-coupling route is independent of the factorization and is statistics-blind: the unperturbed part, electric Casimirs per link together with fermion mass terms, is a commuting, gapped local Hamiltonian with product ground state, and the magnetic and hopping terms are sums of uniformly bounded local perturbations, so uniform-in-volume gap persistence below an explicit threshold follows from the stability theory of gapped Hamiltonians

[45] with or without fermions; making the constants of that theory explicit for the present content is mechanical and remains to be carried out. Finite-volume mass shifts in charged sectors are exponentially small in the volume once the neutral gap and clustering are certified, in the standard manner [46]. Summarizing the reach: at strong coupling, certified infinite-volume gap control is available for all content; at moderate coupling it is available on the stoquastic class conditionally on Open Problem 8.2, and for fermionic content it is open at the structural level (Open Problem 3.6).

8.3 Order

Symmetry-restricted minimization does not move a determined per-volume value: any invariant state is admissible for the restricted problem, and any restricted minimizer averages over the compact symmetry to an invariant state of equal per-volume value (the value being affine and translation covariant). Crystalline or otherwise ordered structure is therefore never an input; whether the optimizing state orders is read from its two-point emissions.

9 Refinement and the continuum

The continuum object is defined relationally: two declared contents are equivalent when all certified reads of $(\mathcal{A}, \omega_0, D_\omega)$ agree; the continuum theory is the equivalence class. This section reduces the existence of that class, for asymptotically free gauge content, to a finite certification problem, and states exactly what remains.

9.1 Refinement is exact

Proposition 9.1. *Budget coarsening acts on moment bodies by exact linear projection: $\mathcal{S}_b$ is the image of $\mathcal{S}_{b'}$ ($b' > b$) under the restriction of moment data. Content refinement (subdivision at declared coupling) composes with this projection through the embedding of the coarser content's algebra in the finer. No approximation enters the map; all dynamical content of the continuum question resides in the behavior of certified brackets along it.*

Proof. Restriction maps $\mathcal{S}_{b'}$ into $\mathcal{S}_b$ because a state's degree- $\leq b'$ data restrict to its degree- $\leq b$ data; surjectivity is immediate because every point of $\mathcal{S}_b$ is by definition the restriction of some admissible state, whose degree- $\leq b'$ data supply a preimage. For composition with content refinement: a state of the finer content restricts along the unital embedding to a state of the coarser, and conversely every state of the coarser content extends to the finer one (positive Hahn–Banach extension in the order-unit norm), so the coarse moment body is the projection of the fine one. $\square$

Remark 9.2 (The relaxations do not project exactly). The corresponding statement for the outer relaxations $\mathcal{M}_b$ of §2.2 is false in general: a level- b feasible moment vector need not extend to a level- b' feasible one. Indeed, if it always did, a diagonal extraction along b' (using the archimedean moment bounds of the compact generators, as in Theorem 9.13) would exhibit every level- b feasible point as the restriction of a genuine state, making every finite level exact and every bracket close at every degree, contradicting Theorem 2.11. The failure of the relaxations to project exactly *is* the tightening of the hierarchy; the exactness claimed by Proposition 9.1 is a property of the bodies, and it is the bodies along which the trajectory of Definition 9.3 is defined.

Definition 9.3 (Trajectory). The *trajectory* is the locus in (content, coupling) along which per-step contraction certificates exist for all emitted relational reads: certified nesting

$$\text{bracket}(n+1) \subseteq \text{bracket}(n),$$

geometric in the directions transverse to the locus, with the declared logarithmic flow along it (the marginal direction of an asymptotically free coupling). The step constraint set at stage n is the certified bracket set emitted at stage $n-1$ together with the content's region constraints; no window parameters exist.

Theorem 9.4 (Tower summation). *If per-step certified control holds along the trajectory with $\delta_n \leq C\rho^n \text{polylog}(n)$, $\rho < 1$, then the certified brackets on every emitted read converge to a limit interval with explicit enclosure $r_N + [-T_N, T_N]$, $T_N = \sum_{n \geq N} \delta_n$.*

Proof. The intervals form a Cauchy sequence in Hausdorff distance with exactly summable tail; the enclosure is the triangle inequality summed from N . $\square$

9.2 Two dimensions are exact

Theorem 9.5. *In two dimensions with heat-kernel plaquette weights*

$$K_t(U) = \sum_j d_j e^{-tC_2(j)} \chi_j(U),$$

subdivision of a plaquette of area t into areas $t_1 + t_2 = t$ leaves every gauge-invariant read exactly invariant.

Proof. Character orthogonality gives $\int \chi_j(UV^\dagger) \chi_k(VW^\dagger) dV = \delta_{jk} \chi_j(UW^\dagger)/d_j$, hence $K_{t_1} * K_{t_2} = K_{t_1+t_2}$: the subdivided theory equals the original on the invariant algebra, and invariant reads depend on enclosed areas alone. $\square$

Theorem 9.5 is the heat-kernel form of the exact solvability of two-dimensional lattice gauge theory [20], transported for the dependency graph; it is the degenerate case of Definition 9.3 (contraction ratio identically zero). At the opposite end of solvability, hierarchical models and free fields constitute the calibration class: certified contraction rates computed there are required to equal the known renormalization-group eigenvalues of those models; rigorous computer-assisted renormalization is an established genre [47, 48], of which the present certificates are the conic instance. A construction that produced rates in disagreement with the known spectra would thereby be refuted as a construction, whatever its internal consistency.

9.3 Slack, dichotomy, and the anchor

Definition 9.6 (Slack). For a state ω , $\sigma(\omega)$ is the supremum, over certificates of the per-step inequality family at fixed degree, of the minimal margin. As a supremum of affine weak-* continuous functionals, σ is weak-* lower semicontinuous and nonnegative on feasible states.

Lemma 9.7 (Uniformity on compacts). *If every state in a weak-* compact set K admits a certificate of degree $\leq b$ with slack $\geq s > 0$, then finitely many certificates cover K and a uniform budget certifies the step on all of K .*

Proof. A fixed certificate's margin is affine and weak-* continuous, so $\text{slack} \geq s$ at a point gives margin $> s/2$ on a neighborhood; compactness (Banach–Alaoglu) extracts a finite subcover. $\square$

Lemma 9.8 (Neighborhood persistence). *If $\sigma(\omega^*) = s^* > 0$ then $\sigma \geq s^*/2$ on a weak-* open neighborhood of ω^* .*

Proof. The certificate achieving s^* has affine continuous margin; its superlevel set at $s^*/2$ is open and lower-bounds σ . $\square$

Theorem 9.9 (Dichotomy). *Along any refinement orbit $\{\omega_n\}$, either $\inf_n \sigma(\omega_n) > 0$, or the orbit has a weak-* accumulation point ω^* with $\sigma(\omega^*) = 0$ exactly.*

Proof. If $\liminf \sigma(\omega_n) = 0$, extract a subsequence with $\sigma \rightarrow 0$; it accumulates by compactness of the state space; lower semicontinuity gives $\sigma(\omega^*) \leq 0$, and nonnegativity closes. $\square$

A zero-slack state is one at which a transverse direction ceases to contract: an unanticipated criticality. The classical failure mode of rigorous renormalization, uncontrolled flow away from the governing fixed point, is here *derived* as the unique failure mode.

The step map factorizes as (variational update) $\circ$ (restriction channel). The channel half is monotone for relative entropy [49], and by the variational formula of [50] relative entropy is itself a supremum of affine functionals of the state, hence an object of the same class as σ ; the variational half-step is where all difficulty resides.

For asymptotically free content the ultraviolet tower flows toward the free fixed point, where the theory is exactly solvable and the slack of the per-step family, with the coupling retained as a bounded parameter, is a finite computation on free content. If that slack is positive, Lemma 9.8 yields an open neighborhood N of the free profile on which a single *parametric* certificate (multipliers polynomial in g over $[0, g^*]$, in the standard manner of [51, 52]) can certify every step, including the forward-invariance inequality keeping the orbit in N ; near the free anchor, effective degree bounds for such certificates [53] render the search decidable with explicit budget. If the free-field slack is not positive at reasonable degree, the strategy of this section fails at its first computation; the question is settled either way by the computation, which has not yet been carried out.

9.4 The reduction

Theorem 9.10 (Reduction). *Suppose (i) a parametric certificate with positive slack and certified forward invariance exists on a neighborhood N of the free profile, uniformly for $g \in [0, g^*]$; and (ii) a finite chain of per-step contraction certificates connects the declared content to N . Then the certified brackets on every emitted relational read converge along the tower with explicit enclosure and uniform budget: the continuum brackets exist, and their existence is verified by finitely many exact-arithmetic identities.*

Proof. The finitely many chain steps carry exhibited certificates; steps beyond the chain are covered by the parametric certificate, whose invariance clause keeps the orbit in N and whose contraction clause supplies δ_n as in Theorem 9.4; that theorem closes the enclosure. The budget is the maximum over a finite set. $\square$

Remark 9.11 (What the reduction changes). Theorem 9.10 changes the *type* of the continuum problem, from a limit statement in analysis to a finite certification, conditional on its hypotheses;

it does not by itself reduce the difficulty, which is relocated intact into hypothesis (ii), the chain of Conjecture 9.16, and into hypothesis (i), whose free-anchor slack computation has not been carried out. The claim that the certificate language has teeth is earned only by executions; the unexecuted decisive computations are named in §12.

9.5 Both branches have witnesses; data cannot decide

Lemma 9.12 (Duality). *On content with compact generators, the per-step feasibility problems at degree $\leq b$ (constraint slices being certified brackets of positive width together with the region constraints) admit infeasibility witnesses: if no certificate exists, there is a normalized functional Λ_b , nonnegative on the truncated quadratic module.*

Proof sketch. The slices have nonempty interior, so the truncated quadratic modules are closed [54, 55]; conic separation yields the witness. For the interior point: the Haar state alone is not feasible for the slices (its moments generally violate the emitted brackets; in the worked instance $\langle x+y+z \rangle_{\text{Haar}} = 0$ against a certified lower bound above 4), but a convex combination $(1-\varepsilon)\omega + \varepsilon\omega_{\text{Haar}}$ of a slice-feasible state with the Haar state is: faithfulness of the Haar state on compact content makes its moment matrices strictly positive definite, so the mixture is strictly interior to the positivity constraints, and the positive width of the bracket slices keeps the mixture inside them for ε small. The Slater alternative then applies. $\square$

Theorem 9.13 (Localization; total dichotomy). *Fixed-degree certifiability is open in the coupling (a certificate with slack absorbs $O(\delta g)$ coefficient perturbations into its margin), so $F_b = \{g \in [0, g_c] : \text{step certificate of degree } \leq b\}$ is open and increasing in b . Consequently either $\bigcup_b F_b = [0, g_c]$ and compactness extracts a finite chain (Theorem 9.10(ii) holds); or the residue $[0, g_c] \setminus \bigcup_b F_b$ is nonempty and compact, and at each of its points the witnesses Λ_b of Lemma 9.12 exist for every b and accumulate weak-* on genuine states of exactly zero slack. Either the chain exists, or a zero-slack obstruction state exists at a localized coupling: both branches of the question have witnesses.*

Proof. Openness: fix a certificate with margin s at g_0 ; the inequality coefficients are polynomial in g , so for $|g - g_0|$ small the same certificate certifies the perturbed inequality with margin $s - O(|g - g_0|)$ by the residual-norm clause of Proposition 2.1; hence F_b is open, and increasing in b because certificates persist under degree enlargement. If the F_b cover $[0, g_c]$, compactness extracts a finite subcover, hence a finite chain. Otherwise fix $g^\dagger$ in the compact residue; for every b a witness Λ_b exists at $g^\dagger$ (Lemma 9.12). Each Λ_b is normalized and positive on squares of degree $\leq b$, and Cauchy–Schwarz on the truncated module together with the exact norm bounds of the compact generators gives $|\Lambda_b(w)| \leq 2^{\deg w}$ for every word of degree $\leq b$. For each fixed degree d , the restrictions of $\{\Lambda_b\}_{b \geq d}$ to the degree- d word space therefore lie in a closed bounded, hence compact, set; a diagonal extraction along b yields a functional Λ on all words, normalized, positive on squares of every degree, with $|\Lambda(w)| \leq 2^{\deg w}$. That bound makes Λ norm-continuous on the dense polynomial subalgebra, so it extends to a state on the C^* closure. Finally, each Λ_b assigns the step inequality a nonpositive value while satisfying every slice constraint (that is the witness property); the finitely many margins are jointly continuous in (Λ, g) , so the limit state assigns the step inequality a nonpositive value while lying in the constraint slice: a state of exactly zero slack at $g^\dagger$, since positive slack at the limit would, by Lemma 9.8, certify a weak-* neighborhood meeting the Λ_b and contradict their witness property. $\square$

Theorem 9.14 (No decision by data). *The witnesses Λ_b satisfy every slice constraint by construction (hence every certified bracket the instrument has emitted or will emit on the declared reads, up to their degree), and so does any limiting obstruction state. Consequently no finite accumulation of certified brackets on the physical state entails the nonexistence of the obstruction: the branches of Theorem 9.13 cannot be separated by bracket data alone. They can be separated only by exhibition of the chain, or by a structural theorem excluding the obstruction.*

Remark 9.15 (What failure looks like). By Lemma 9.12, failure at fixed degree is itself decidable and carries a witness; a sustained failure across degrees produces a sequence of witness functionals converging on the obstruction’s own moment profile, which can be symmetrized and interrogated by every declared read. Unsuccessful search at bounded degree is therefore informative; unbounded non-existence is not finitely refutable, which is the precise character of the remaining problem.

9.6 The central conjecture and candidate mechanisms

Conjecture 9.16 (Existence of the chain). *For asymptotically free gauge content in 3+1 dimensions, the finite chain of Theorem 9.10(ii) exists. Equivalently (Theorem 9.13): no zero-slack obstruction state lies on the segment between the declared content and the free anchor. This is nonperturbative asymptotic freedom in certificate form; upon exhibition of the chain, the continuum brackets of the theory exist at the strength of finitely many verified identities.*

Three mechanisms, each with a decisive computation in the calibration class, would produce or organize the chain. (a) *Curvature condition*. The canonical per-step certificate is the running state’s own geometry: transverse contraction as a uniform spectral gap of the state Dirichlet forms, certified by a curvature-dimension inequality $\Gamma_2 \geq \kappa \Gamma$ [56] in the exact Γ -calculus of §3.4; whether the per-step certificates of the solvable controls can be taken in this form is a computation. On fermionic content this mechanism inherits the scope boundary of Remark 3.5 and Open Problem 3.6. (b) *Monotone*. A certified functional of the state, in the affine-supremum class of Definition 9.6, strictly decreasing along the refinement flow with quantitative strictness away from the free fixed point, would exclude the obstruction branch outright, and with the free anchor computation would imply Conjecture 9.16 by Theorem 9.13; the entropic proofs of renormalization-flow monotonicity [57] indicate the class in which to look, and the requirement here, a monotone for one flow on one content class, is strictly weaker than a universal theorem. (c) *Parametric collapse*. If the chain’s certificates can be taken with coefficients polynomial in g over the whole segment, the chain is a single parametric certificate; whether the per-step certificates of the controls vary polynomially along the flow is a measurement.

10 Hadronic content

10.1 Fermions at declared content

Quark content enters as CAR sectors coupled to the gauge content, the Gauss law imposed by exact quotient on the invariant algebra (at declared lattice content the quotient is exact and requires no section, so the obstruction that blocks continuum nonabelian gauge fixing does not arise at this level). Lattice chirality is constrained: no discretization carries exact chiral symmetry, doubler freedom, and locality simultaneously [58]; the class defined by the Ginsparg–Wilson relation $\{\gamma_5, D\} = aD\gamma_5D$ [59, 60] carries the exact algebraic form of chirality and is

adopted as content. Residual parameters within the class are declared, and every emission is required to be stable across the declared parameter window at its stated resolution; parameter dependence at resolution refutes the emission, not the framework.

10.2 Spectral hadrons

Baryon and flavor numbers are canonical gradings of the content (§3.4); the mass of the lightest baryon is the ground-sector energy difference between grading sectors,

$$m_B = E_0(B=1) - E_0(B=0),$$

a spectral quantity requiring no interpolating operators, no plateau analysis, and no statistical estimator: both terms are certified brackets, and the difference inherits interval arithmetic. Emissions are dimensionless ratios (e.g. m_B/m_M against the lightest meson); each content parameter (quark-mass coupling) is fixed by exactly one declared calibration ratio, disjoint from all comparison targets, and electromagnetic and isospin-breaking effects are outside the stated resolution tier until declared otherwise.

10.3 What a hadronic emission requires

A certified m_B/m_M at declared lattice content and coupling requires: the fermionic instance of the canonical machinery (Open Problem 7.4); certified brackets in two grading sectors on volumes governed by §8, where the reach is exactly as summarized in §8.2: Theorem 8.1 is statistics-blind (fermionic hopping terms crossing a tile boundary are norm-bounded per link and mass terms partition, so the proof extends verbatim), the strong-coupling gap control is statistics-blind, and the moderate-coupling gap criterion does *not* reach fermionic content (Open Problem 3.6); and, for the comparison to carry weight, the strong-coupling value of the ratio derived independently in advance, with the resolution rule of §2.6 in force. The strong-coupling lattice baryon program is therefore reachable by the statistics-blind route; away from strong coupling, certified fermionic gap control is open at the structural level. A certified *continuum* m_B/m_M , the proton-to-meson ratio of the physical theory, requires in addition exactly the content of §9: it exists upon Conjecture 9.16 together with the free-anchor computation, by Theorem 9.10, and by Theorem 9.14 no accumulation of lattice data can substitute for that exhibition. The open items between the present state of the framework and a physical proton mass are therefore precisely: Open Problems 7.4, 8.2, and 3.6, the explicit constants of §8.2, the free-anchor computation, and Conjecture 9.16. These are not of one kind: all but one are completed by work of a known character, while Open Problem 3.6, the sign problem in carrier form, may be a structural obstruction rather than an unfinished task, and any account of the path that omits it, or lists it as routine, misstates the theory's position. Nothing else stands in the path, and each item is stated where its mathematics lives.

11 Electrodynamic matter

This section declares the content of electrodynamic matter, with gravitation carried as a spin-2 boson sector, and develops it inside the framework. Historically this instance was constructed first; the general theory of §§2–9 is its closure, and everything below is that theory evaluated at declared content.

11.1 Content

The arena is $\mathbb{R}^3$, entering the theory exclusively through the coefficient data of the local dynamics (kernels, dispersions, form factors); this is the entire residual background. A system is declared by its composition: the species present, a closed subgroup $H \subseteq \mathbb{R}^3$ (the *content group*; every closed subgroup is $V \oplus \Lambda$ with V a subspace and Λ a lattice in a complement) under which the matter distribution is homogeneous, and the species densities on $\mathbb{R}^3/H$: bulk matter is $H = \mathbb{R}^3$, a molecule $H = \{0\}$. The content group declares what system is studied, never how its matter organizes; no lattice, cell, or space group is among the data (§8.3).

Species are second-quantized fields. Per charged-lepton flavor, the CAR algebra over

$$L^2(\mathbb{R}^3) \otimes \mathbb{C}^4$$

graded by electric charge: particles and antiparticles carried together, particle number not conserved, flavor charge conserved, and *no positive-energy subspace selected at any point for any purpose* (§11.4). Per declared isotope, a CAR or Weyl-form CCR field over $L^2(\mathbb{R}^3) \otimes \mathbb{C}^{2s+1}$ with mass, charge, spin, and charge form factor as Hamiltonian data of the standing of the mass. The photon is the transverse Weyl algebra at bandwidth (two polarizations). The graviton is the transverse-traceless spin-2 Weyl sector, coupled to the matter stress words exactly as the photon couples to the charge current, with $\kappa = \sqrt{16\pi G}$ a measured datum of the standing of a charge; no separate gravitational theory is carried. Statistics are algebra identities; no symmetrization machinery exists. Admissible states are positive, normalized, H -invariant, invariant under each flavor's charge $U(1)$ at the declared densities, neutral in total, with finite boson second moments; per-covolume quantities live in the density quotient of local elements modulo translates. The Hamiltonian density assembles the Dirac coefficient matrices, the isotope dispersions, the instantaneous Coulomb interaction of the *total* charge density (produced by the quotient below), the transverse current coupling, the free field energies at budget, and the transverse-traceless stress coupling.

Proposition 11.1 (Boundedness below at budget). *At every stated budget, $\hat{h} \geq -C(\text{budget}) \mathbf{1}$ as a form on the admissible class.*

Proof. Fermionic smeared words with bounded coefficients are bounded; kinetic and boson number forms are nonnegative; the reciprocal Coulomb part is a positive superposition of Hermitian squares of smeared total-charge words; the singular slice of the point-charge specialization is form-dominated by a declared fraction of the lepton kinetic bilinear (Hardy), itself bounded at bandwidth; and boson-linear couplings obey, mode by mode, $\pm(Ba^\dagger + B^\dagger a) \leq \varepsilon\omega a^\dagger a + (\varepsilon\omega)^{-1}B^\dagger B$ with B bounded, the first term absorbed into a fraction of the free energy over the budgeted band. $\square$

Proposition 11.2 (Two-sided transfer of kinetic truncation). *If $\hat{h}^{(t)}$ carries, for any subset of massive species, kinetic dispersions truncated from below ($0 \leq t(k) \leq k^2/2m$ pointwise), then $\hat{h}^{(t)} \leq \hat{h}$ as forms: every certified lower bound for the truncated theory transfers, while every exhibited state evaluates the full dispersions exactly. Kinetic truncation costs width only, on both sides, never soundness.*

Proposition 11.3 (The photon axis is variational from above). *e_0 is monotone nonincreasing in the photon budget: a budgeted admissible state extended by the vacuum on the new band evaluates the enlarged Hamiltonian to the same energy density. Enlarging the budget enlarges the coupling, which is indefinite, so no form inequality exists; lower bounds are per-budget, and the behavior of anchored relational emissions along the axis is the measured trajectory of §11.9.*

11.2 The exact abelian quotient

The electromagnetic phase space is reduced, never gauge fixed: the abelian Gauss constraint is solved exactly on $\mathbb{R}^3$, the longitudinal electric field is determined by the charge density, its elimination produces the instantaneous Coulomb interaction, and the remaining local degrees of freedom are the two transverse polarizations. This is a quotient with nothing selected; at $\mathbf{H} = \mathbb{R}^3$ or $\{0\}$ there are no cycles, hence no holonomies and no flux labels. It is the $G = U(1)$ case of the exact quotient of §2.1.

Remark 11.4 (Torus holonomies at periodic content). At a content group $\mathbf{H} = V \oplus \Lambda$ with a nontrivial lattice Λ of rank $d \geq 1$ the reduced configuration space carries, besides the local transverse modes, the finitely many global harmonic modes of the periodic geometry: the holonomies of the connection around the d cycles with their conjugate momenta, and at $d \geq 2$ integer magnetic flux labels through the 2-cycles. Both dispositions are exact and cost nothing. The flux labels are superselected sector data, declared with the particle content; the holonomy pairs are a fixed finite set of global Weyl generators whose contribution to every per-covolume density vanishes in the quotient, so no emission sees them until a flux-threading emission is wanted, when they are adjoined explicitly. The blanket absence above holds only at the two endpoint content groups.

Remark 11.5 (Constraints without sections). Where a constraint's solution space is not a subalgebra (the nonabelian gauge orbit, which admits no global section), this framework forms no projection at all: content is declared on the invariant algebra directly, as in §§5, 10, and the constrained non-subalgebra projection never arises as an object of the theory. The continuum question for the nonabelian sectors is then exactly Conjecture 9.16.

11.3 The physical inner product and its reads

Definition 11.6 (Single emission of the instance). The *physical inner product* of the content is $\langle A, B \rangle = \omega_0(A^\dagger B)$ on the word spaces: the GNS form of §3 and the moment functional of §2.2, carried projectively (the state normalization is gauge). Every certified number of the instance is a two-sided bracket on a matrix element of this form over the ground sector, taken directly or through a conditional expectation onto a translation-covariant subalgebra (which preserves both positivity and the per-covolume average, and is the only conditioning class admitted).

The energy axis carries an affine gauge, the additive zero point and the multiplicative clock rate, whose invariants are exactly the ratios of sector differences. The absolute energy is evaluation-internal; every emitted physical quantity is such a ratio, with the reference transition a measured datum that doubles as a calibration anchor. The remaining reads: sector differences at distinct declared compositions; the structure factor $\omega_0(\hat{\rho}(k)^\dagger \hat{\rho}(k))$, whose Bragg atoms are crystalline order read and never imposed (§8.3); the frame, read from the vanishing of the momentum-density one-point; modular time, the admissible-subalgebra modular structure of §3.2, exactly evaluable for quasi-free and Gaussian restrictions under a full-rank condition on the restricted covariance that fails shut on degeneracy; and the response family, the jet of the certified energy tube under source couplings $\hat{h} \mapsto \hat{h} + \sum_\alpha \theta_\alpha O_\alpha$: the gradient is the one-point read (order parameters and frame; Hellmann–Feynman at stationarity, envelope one-sided derivatives in general, degenerate optima failing shut), the Hessian the static susceptibilities, and time-dependent sources the complex-time two-point whose analyticity is the modular read.

11.4 The polarization is an output

The first-quantized Dirac–Coulomb many-body problem is unbounded below [68]. Confronted with it, a two-sided instrument fails shut: no certificate exists, which is the true statement, and variational collapse (a disease of one-sided methods) cannot occur. On the charge-graded field algebra no positive-energy subspace is ever selected: the polarization, the projector that no-pair models fix by hand in representation-dependent ways, is here the level-one moment datum of the state, an output on the same footing as the nuclear covariance (phonons), the photon covariance and displacement (the classical field), and the Bragg content (the crystal). These are the level-one reads of the reference rung (Proposition 2.5) at the field group, the quasi-free $\otimes$ Gaussian data the singlet fusion determines, with the modular generator (the flow) the fifth. The stability literature corroborates the direction: the no-pair model is unstable with the bare-Dirac electron subspace [69] and stable with the field-dressed one [70]; the physically correct polarization was never a prior selection but a property of the interacting state, which is where the physical inner product places it. Ground states of the nonrelativistic quantized-field models exist at all coupling [71], supplying the exhibited-state side of the brackets.

11.5 Specializations and calibration

Four theories, each a coefficient substitution with a declared budget restriction in the one above, never an approximation within the finer theory: full quantum electrodynamics at cutoff; the Pauli–Fierz theory (pair sectors frozen, nonrelativistic kinetic coefficients); the Coulomb theory (photon budget empty); the clamped theory (isotopes replaced by classical point sources at parametric positions). Certified intervals of adjacent theories at shared budgets bracket, telescopically, the nuclear-motion content, the radiative and retardation content, and the relativistic and pair content; no endpoint is estimated, and in relational form the brackets remain certified and dimensionless. Isotope shifts decompose into certified field and motional parts, whose reversal reads nuclear charge radii. Renormalization is budget-running data: one measured anchor per counterterm, no anchor anywhere else, each bare parameter an interval-valued datum propagated by interval arithmetic; calibration costs width, never soundness.

Proposition 11.7 (Certified approximation brackets). *Certified intervals of adjacent theories at shared budgets bracket, no endpoint estimated: the nuclear-motion content in $[L_{\text{Coul}} - U_{\text{Cl}}, U_{\text{Coul}} - L_{\text{Cl}}]$, the radiative and retardation content in $[L_{\text{PF}} - U_{\text{Coul}}, U_{\text{PF}} - L_{\text{Coul}}]$, and the relativistic and pair content in $[L_{\text{QED}} - U_{\text{PF}}, U_{\text{QED}} - L_{\text{PF}}]$. The brackets compose telescopically, and in relational form they divide by the reference charge and stay certified and dimensionless. Isotope structure sharpens along the ladder: clamped energies carry the form factors but not the masses, so their isotope differences are certified field shifts; joint emissions carry both, so certified isotope shifts decompose into field and motional parts, the reversal reading nuclear charge radii, with the radiative bracket the muonic-hydrogen read.*

11.6 Convergence for this content

Theorem 11.8 (Finite range). *For $\mathbf{H}$ -covariant Hamiltonian densities of finite range satisfying Proposition 11.1, the certificate hierarchy converges: $\sup_{\beta} L^* = e_0$.*

Proof outline. Tile $\mathbf{H}$ by overlapping weighted boxes; on each box the shifted cluster Hamiltonian, tilted by density multipliers, is a positive element, hence a Hermitian square whose factor is approximable by cluster words with certified remainder; cluster grand energies converge per

covolume, and Legendre duality with convexity of the energy density in the densities, known for the Coulomb class [72], closes the argument. The functional-analytic control of the square-root approximation on the infinite-dimensional bosonic box sectors is the one step not carried out here (Open Problem 11.13). $\square$

For the instantaneous interaction, the identity of Fefferman and de la Llave [73],

$$\frac{1}{|x - y|} = \frac{1}{\pi} \int_0^\infty \frac{dr}{r^5} \int_{\mathbb{R}^3} dz \mathbf{1}_{B(z,r)}(x) \mathbf{1}_{B(z,r)}(y),$$

decomposes the total charge–charge interaction exactly into a finite-range singular part ($r < r_0$, absorbed into the cluster energies), a bounded one-body remainder, and for $r \geq r_0$ a part manifestly a positive superposition of Hermitian squares of smeared *total* ball-charge words $\hat{Q}_B$, species-blind by construction, every species a field density. With a spectral guard $\hat{G}(k) \geq g(r_0)$ on the low modes $k < 1/r_0$ (with $g = c_3 r_0^2$, $c_2 r_0$, $c_1 \ln r_0$ by noncompact dimension 3, 2, 1), guard manipulations stay in the certificate class.

Lemma 11.9 (Mass bound). *For every admissible ω within unit energy of the remainder ground, $\int_{k < 1/r_0} \hat{G} d\hat{C}_\omega \leq C_0$, hence $\int_{k < 1/r_0} d\hat{C}_\omega \leq C_0/g$: the low-mode charge fluctuation the guard admits is $O(1/g)$.*

Lemma 11.10 (Phase-space integrability; kernel–measure). *With $W(k)$ the long-wavelength kernel, $W(k) dk$ is bounded on the guarded ball up to a logarithm at noncompact dimension one. The $k = 0$ mode is the exactly conserved total charge: at $\mathbf{H} = \{0\}$ sector sharpness kills its atom outright, and at noncompact $\mathbf{H}$ the atom is removed by a conditioning step at its degenerate-best instance (identically vanishing f -sum), carrying the status of Theorem 11.11.*

Theorem 11.11 (Cost Theorem). *There are constants K_1, K_2 from budget data alone (local operator norms, the reference f -sum $\sum_s n_s q_s^2/m_s$ over charged species, the stripped response constant A) such that for every admissible ω within unit energy of the remainder ground there is an admissible $\tilde{\omega}$ with remainder expectation at most $K_1 g^{-1}$ larger and deleted-part expectation at most $K_2 g^{-1}$. Hence $\Delta_{\text{reopt}}(r_0) \leq K g(r_0)^{-1} \rightarrow 0$, and with Theorem 11.8 on the finite-range remainder, $\sup_\beta L^* = e_0$ at every content group, gapped or metallic, ordered or not.*

Proof outline. Every object is a functional of the *total* charge: dyadic shells with smeared total-charge quadratures and reference conjugates, one linear cost law

$$\Delta E_j \leq \min(\text{squeeze}, \text{condition}) + A(S - B),$$

iterated squeezing with re-derived mass bounds (Lemma 11.9), median-halving conditioning where the shell f -sum is depleted, outer-to-inner sweeps, $B \sim r_0^{-2}$. The enumerated items are Open Problem 11.13, and each mechanism is pinned in a closed-form solvable model in Appendix C. $\square$

Content geometries. At $\mathbf{H} = \{0\}$ flavor-charge sharpness strengthens the certificate class: sharp states kill the full one-sided ideal $(\hat{Q}_\ell - q_\ell)\mathcal{A} + \mathcal{A}(\hat{Q}_\ell - q_\ell)$, since $(\hat{Q}_\ell - q_\ell)\Omega = 0$ in the carrier. For noncompact $\mathbf{H}$ the constraints are density expectations, sharp nowhere, and only the scalar multipliers are killed; *admitting the ideal there is unsound*. This is the one point at which content geometries differ in the certificate class.

Proposition 11.12 (Convergence at noncompact dimension 1, 2). $\sup_{\beta} L^* = e_0$ at $H = \mathbb{R}, \mathbb{R}^2$ by the proof of Theorem 11.11 with the substitutions: tiling and thermodynamic limit over H ; the Fefferman–de la Llave decomposition unchanged (it decomposes the interaction in $\mathbb{R}^3$, independent of the content group); the diverging guard saturations $g = c_2 r_0, c_1 \ln r_0$; and for polar slabs the slab form of the kernel, the transverse direction an integral, the dipole term finite and exact. Rates degrade with decreasing dimension.

Open Problem 11.13. The enumerated items of Theorems 11.8 and 11.11: the bosonic-sector square-root approximation; the many-body second-order bookkeeping and explicit quasi-locality constants of the cost law; the $k = 0$ conditioning step at noncompact content groups (Lemma 11.10); the carry-over to the boson-linear (transverse) couplings, per frequency shell with the photon frequency the guard’s analogue; and the low-dimension constants and slab step of Proposition 11.12. Each is functional-analytic; none affects the certified per-budget soundness of any emission.

Theorem 11.14 (Clamped exactness at $H = \{0\}$). *In the clamped theory with m budgeted electronic modes, the bracket closes at word degree $2m$: $L^* = e_0 = U^*$.*

Proof. Monomials of degree $\leq 2m$ in the mode operators are a linear basis of the CAR algebra (which is 4^m -dimensional), so the degree- $2m$ word space is the whole algebra: the moment data at this budget are the functional itself, and Gram positivity over a spanning word set is positivity on every Hermitian square, i.e. the moment cone at this degree *is* the state set. With $P_{N'}$ the number-sector projectors (themselves words of degree $\leq 2m$), $\hat{H} - \sum_{N'} E_0(N') P_{N'} \succeq 0$ is a sum of Hermitian squares of algebra elements, hence a certificate at this degree, giving the lower value $E_0(N)$ on the declared sector; the exhibited side attains it at full richness. $\square$

Remark 11.15 (The half-degree relaxation). Products of two degree- $\leq m$ words already span the algebra, so the level- m relaxation determines every moment of the functional. Whether its positivity constraints alone are already exact (closure at word degree m rather than $2m$) is a positivity-transfer question: determination of moments does not by itself imply exactness of a relaxation, and positivity on the squares of a spanning-product subspace is strictly weaker than full positivity in general. It is verified where used and not claimed here.

11.7 Gravitational reads

Every measured gravitational quantity is a read of the physical inner product; none is an addition to the machinery. The sector’s structure must be stated precisely, because a single adjective otherwise does invisible work.

Definition 11.16 (The gravitational source, exactly). The transverse-traceless spin-2 sector couples *linearly* to the matter stress words. The longitudinal sector is not a degree of freedom: it is fixed by its source through the gravitational Gauss law, exactly as the Coulomb field is fixed by the charge, and the source of the constraint is the total energy density *including the field’s own*, carried exactly once by the coordinate-measure mass function $m(r) = \int_0^r 4\pi r'^2 \rho dr'$: the field self-energy enters through the *measure* (coordinate volume in place of proper volume), their difference being the gravitational binding energy exactly. No iterated dynamical self-stress vertex is declared, and no nonlinear field equation is imported. The clamped metric read is $g_{00} = 1 - 2Gm(r)/r$ with $g_{rr} = 1/g_{00}$; the linearity is in the variable m , not a linearization of anything.

Remark 11.17 (Relation to the self-coupling bootstrap). A reader trained on the Fierz–Pauli-to-Einstein bootstrap [61, 62] will ask whether Definition 11.16 is that construction. It is not, and the difference matters in both directions. The bootstrap iterates a coupling of the linear field to a total stress that includes the field’s own; its conclusion is frequently cited as an unconditional identity between linear spin-2 and general relativity, but the iteration requires a choice of gravitational stress tensor at every step (superpotential and field-redefinition freedom), and with that freedom the bootstrap is a construction with inputs, not an identity [63, 64]. The present framework does not perform the iteration and does not inherit the ambiguity: the field’s self-energy enters the source exactly once, through the solved constraint of Definition 11.16, fixed by the same quotient that fixes the Coulomb interaction: there is no stress tensor to choose. The comparison with general relativity is therefore made emission by emission, and the first place the two constructions could differ on a purely two-body observable is stated below as an open problem rather than absorbed into a bootstrap citation.

The ledger of the comparison, at the orders measured. *Test-mass sector*: matter geodesics in the clamped read metric carry the perihelion advance, light deflection, gravitational redshift, and Shapiro delay at their measured orders. Frame dragging is not carried by that metric, which is static and diagonal and has no g_{0i} ; it is carried by the linear mass-current (T^{0i}) coupling of the same spin-2 sector, the channel that also supplies the retardation term below, and linear theory reproduces the Lense–Thirring precession at measured accuracy. *Two-body first post-Newtonian order*: the velocity structure of the Einstein–Infeld–Hoffmann dynamics [65] is derived within the sector as declared: the instantaneous exchange read gives $(Gm_1m_2/2r) [3(v_1^2+v_2^2) - 8v_1\cdot v_2]$, and the retardation kernel supplies exactly $(Gm_1m_2/2r) [v_1\cdot v_2 - (n\cdot v_1)(n\cdot v_2)]$, summing to the EIH form $(Gm_1m_2/2r) [3(v_1^2+v_2^2) - 7v_1\cdot v_2 - (n\cdot v_1)(n\cdot v_2)]$. The computation is half a page and is given in Appendix B, together with its abelian control: the identical kernel identity evaluated in the $U(1)$ sector yields the Darwin Lagrangian [67], a known result. *Two-body second post-Newtonian order*: open, and stated as such.

Open Problem 11.18 (The first purely two-body self-source observable). At second post-Newtonian order the two-body dynamics first carries a coefficient, linear in the symmetric mass ratio ν , that no test-mass read and no constraint normalization determines: the -7ν of the known two-body dynamics [66]. Derive it within the sector as declared, from the forced-source structure of Definition 11.16. Until this derivation exists, the gravitational sector’s agreement with general relativity is *derived* through first post-Newtonian order and the test-mass series, and *required but not derived* at second post-Newtonian two-body order; a derivation producing any value other than the known one refutes the sector as declared. The standard expectation is that a linear field with a solved constraint fails at precisely this coefficient; the framework does not dispute where the burden lies and accepts the exposure as stated. The problem’s scope includes the mass function of Definition 11.16: that the solved constraint carries the binding energy through the measure is derived at the leading order the comparisons above use, and asserted beyond it.

Radiation reads are unchanged by the above: the binary-pulsar orbital decay is the certified emission bracket of the spin-2 sector; a coalescence is read from its radiation alone (radiated energy and angular momentum are certified quadratic brackets on the emitted transverse-traceless field, the final mass and spin are conservation reads, the ringdown is the linear spectral read of the final field). Because the interaction is attractive, the free-mass source problem is unbounded below and the instrument fails shut on it; gravitational reads enter through the clamped specialization, as they must for a self-attracting source.

11.8 An executed control

The certified nonrelativistic hydrogen frequency ratio $f(1S-3S)/f(1S-2S)$, emitted with certified width 2.9×10^{-12} , excludes the measured value by $+1.309 \times 10^{-6}$; the exclusion is quantitatively the uncarried relativistic sector: the Dirac correction $+1.109 \times 10^{-6}$ plus the Lamb-shift differential. What this exercise establishes, exactly: the emission pipeline (interval arithmetic, the relational calibration-free form of the ratio, and the exclusion semantics by which the specialization brackets of the ladder are read) operates end to end on a case where both numbers are independently known. What it does not establish: the ratio at this content is Rydberg-exact, so no moment-cone certificate is exercised, and no claim of the theory is tested that was not already known analytically. In the paper's own vocabulary this is a control, not a validation; the nontrivial certificate machinery is exercised where the instance is not exactly solvable, as in §5.

11.9 The ultraviolet axis

Every theorem of this section holds at stated budgets. Whether certificates exist whose constants, with the calibration anchors re-solved at each ultraviolet budget, are uniform in the budget is the refinement problem of §9 evaluated at this content: renormalized limits of exactly this kind are proved for the Nelson model [74], cutoff-uniform stability constants are proved for nonrelativistic quantized-field matter [75], and the scalar triviality results [76] neither establish nor preclude anything here. Along the axis the anchored relational emissions are certified intervals, so convergence or drift of the renormalized family is measured data; cutoff insensitivity is, on this instrument, a number with a proof, or it is false. The abelian ultraviolet axis and the nonabelian chain of Conjecture 9.16 are one problem, the existence of certified contraction along refinement, posed at two declared contents.

12 Concluding remarks

The theory is a constructor: from declared content it builds the state, and from the state its time, geometry, order, thermodynamics, and, conditionally, its continuum, each object canonical, each emission a verified two-sided bracket, each unproven step a named statement. The proved core comprises the kinematics and soundness results (§2), the canonical constructions with the factorization and harmonicity theorems (§3), the density theorem (§8), the exactness of refinement and of two dimensions, the dichotomy, localization, no-go, and reduction theorems (§9), and the electrodynamic-matter theorems of §11. Proof statuses are graded in place: complete proofs (the densities, tower summation, two dimensions, uniformity, persistence, dichotomy, localization, no-decision-by-data, clamped exactness, and the derivations of the appendices); complete proofs under stated domain hypotheses (harmonicity, factorization); and outlines with their deferred steps named (the finite-range embedding; the closedness import in the duality lemma). The abstract claims completeness only where it holds. The open surface comprises: the fermionic machinery of Open Problem 7.4; the carrier transport of Open Problem 8.2 and the explicit stability constants of §8.2; the local factorization for fermionic content of Open Problem 3.6, which unlike the others may be a structural obstruction rather than an unfinished construction; the dual-pair presentation of Open Problem 6.4, with Conjecture 6.8 its candidate criterion for that boundary; the convergence completions of Open Problem 11.13; the second post-Newtonian two-body derivation of Open Problem 11.18; Hypothesis 7.3 with its four decisive free-fermion

tests and the signature-consistency requirement they also carry (Definition 7.5); the free-anchor slack computation of §9; and Conjecture 9.16, whose statement Theorems 9.13 and 9.14 fence exactly: both branches carry witnesses, no accumulation of certified data decides between them, and resolution comes only by exhibition of a finite verified object or by a structural monotone. One decisive computation the framework imposed on itself has been executed and passed: the Weyl consistency comparison of Definition 4.1 on the worked instance closes exactly, both sides in closed form (Theorem 4.2; Appendix A.4), the first decisive test of the geometric identification, passed at the strength of an algebraic identity. Four remain unexecuted, and are named as such: the prepotential re-pin of §6, the certified brackets of the worked instance re-derived in dual-pair words with digit-level agreement required; the free-anchor slack computation of §9; the four free-fermion tests of Hypothesis 7.3, blocked on Open Problem 7.4; and the signature-consistency comparison of Definition 7.5, downstream of the same tests, on which the real form of the spacetime group is an output. A further grading concerns provenance: several labeled theorems are standard facts transported into this frame and cited as such (the density theorem is a subadditivity argument; the canonical stack recombines the GNS, Tomita–Takesaki, and Cipriani–Sauvageot constructions; the fusion and closure propositions of §6 are classical invariant theory and Howe duality); they are numbered for the dependency graph, not claimed as new. What the paper offers as its own are the resolving-emission and exclusion discipline, the factorization scope with the sign problem in carrier form, the exactness of refinement, the total dichotomy with its no-decision corollary, the reduction theorem, the field–gauge correspondence read as group-from-closure with its dual-pair criterion and its spacetime reading (complex point from the ideal, real form from the modular conjugations), and the quotient-fixed gravitational reads with their stated exposure. The calibration requirements of §9 likewise bind; the hadronic targets of §10 assemble from the theorems and open statements above with nothing additional, which is the intended sense in which this document presents one theory rather than a collection of results.

A Derivations for the worked instance

Charts appear below as evaluation scaffolding for chart-free quantities (a Hausdorff volume, a distance, a spectral expansion); they enter no definition and no emission.

A.1 The metric

On one copy of $SU(2)$ write $U = \cos \theta + i \sin \theta \hat{n} \cdot \sigma$, $\theta \in [0, \pi]$, $\hat{n} \in S^2$. Bi-invariance forces the round form $ds^2 = R^2[d\theta^2 + \sin^2 \theta |d\hat{n}|^2]$, and the normalization follows from the verified carré du champ: with $x = 2 \cos \theta$, $\Gamma(x, x) = g^{\theta\theta}(\partial_\theta x)^2 = g^{\theta\theta} 4 \sin^2 \theta$ must equal $(4 - x^2)/4 = \sin^2 \theta$, so $g^{\theta\theta} = 1/4$, i.e. $g_{\theta\theta} = R^2 = 4$. On the invariant sector of the pair, use coordinates $(\theta_1, \theta_2, \beta)$ with β the angle between the two axes. Then $|\nabla \theta_i|^2 = 1/4$; the axis sphere of copy i has radius $2 \sin \theta_i$, so moving axis i changes β at rate $|\nabla^{(i)} \beta|^2 = 1/(4 \sin^2 \theta_i)$; and the θ - and β -gradients are mutually orthogonal (distinct tensor factors; radial versus tangential on each factor). Hence

$$g^{\beta\beta} = \frac{s_1^2 + s_2^2}{4 s_1^2 s_2^2}, \quad g = \text{diag}\left(4, 4, \frac{4 s_1^2 s_2^2}{s_1^2 + s_2^2}\right), \quad \sqrt{\det g} = \frac{8 s_1 s_2}{\sqrt{s_1^2 + s_2^2}},$$

with $s_i = \sin \theta_i$.

A.2 The volume

$$\text{Vol} = \int_0^\pi d\beta \int_0^\pi \int_0^\pi \frac{8 s_1 s_2}{\sqrt{s_1^2 + s_2^2}} d\theta_1 d\theta_2 = 8\pi \int_{[-1,1]^2} \frac{du dv}{\sqrt{2 - u^2 - v^2}}$$

(substituting $u = \cos \theta_1, v = \cos \theta_2$). Split the square into the inscribed unit disk and four corner regions. Disk: $\int_0^{2\pi} \int_0^1 \rho d\rho d\varphi (2 - \rho^2)^{-1/2} = 2\pi(\sqrt{2} - 1)$. Corner (first quadrant, $u^2 + v^2 \geq 1$): in polar form $J = 2 \int_0^{\pi/4} [1 - \sqrt{\cos 2\varphi} / \cos \varphi] d\varphi$; the substitutions $s = \sin \varphi$, then $s = \sin \psi / \sqrt{2}$, give $\int_0^{\pi/4} \sqrt{\cos 2\varphi} / \cos \varphi d\varphi = \sqrt{2} \int_0^{\pi/2} \cos^2 \psi / (1 + \cos^2 \psi) d\psi = (\pi/2)(\sqrt{2} - 1)$, hence $J = \pi(3/2 - \sqrt{2})$. The square integral is $2\pi(\sqrt{2} - 1) + 4\pi(3/2 - \sqrt{2}) = (4 - 2\sqrt{2})\pi$, and

$$\text{Vol} = 8\pi \cdot (4 - 2\sqrt{2})\pi = (32 - 16\sqrt{2})\pi^2.$$

A.3 Distances

One loop: class functions of a single copy carry the metric $4d\theta^2$, so

$$d = 2|\theta_1 - \theta_2| = 2|\arccos(x_1/2) - \arccos(x_2/2)|.$$

Diameter of the two-loop variety: the projection $(\theta_1, \theta_2, \beta) \mapsto (\theta_1, \theta_2)$ is metric-nonincreasing onto the flat square with metric $4(d\theta_1^2 + d\theta_2^2)$, giving

$$d(\mathbf{1}, (-\mathbf{1}, -\mathbf{1})) \geq 2\sqrt{\pi^2 + \pi^2} = 2\sqrt{2}\pi;$$

the constant- β diagonal path attains this length, so equality holds. That this pair realizes the diameter follows from the same comparison run upward: the invariant sector is a metric quotient of $S^3(2) \times S^3(2)$ (radius-2 round factors), quotient maps are distance-nonincreasing, and the product's diameter is

$$\sqrt{(2\pi)^2 + (2\pi)^2} = 2\sqrt{2}\pi,$$

so no pair is farther.

A.4 The spectral side and Weyl consistency

This subsection proves Theorem 4.2. The general statement behind it is classical: the equivariant Weyl law of Donnelly and Brüning–Heintze [17, 18], by which the counting function of invariant eigenfunctions is governed by the volume and dimension of the orbit space. What is verified here is therefore not the law but the framework: every normalization on both sides was fixed elsewhere, and the instance is computed directly in closed form. The carrier at the reference is the closure of the invariant algebra in $L^2(\text{SU}(2)^2, \text{Haar})$: the simultaneous-conjugation-invariant sector.

Multiplicities. Peter–Weyl gives $L^2(\text{SU}(2)) = \bigoplus_j V_j \otimes V_j^*$, and conjugation acts on the j block as the adjoint-type action on $\text{End}(V_j) \cong V_j \otimes V_j^* \cong \bigoplus_{l=0}^{2j} V_l$ (integer l). For the pair, the diagonal-conjugation invariants of the (j_1, j_2) block number $\dim \text{Inv}(\bigoplus_{l_1} V_{l_1} \otimes \bigoplus_{l_2} V_{l_2}) = \#\{l : l \leq \min(2j_1, 2j_2)\} = \min(2j_1, 2j_2) + 1$, at Casimir eigenvalue $j_1(j_1+1) + j_2(j_2+1)$. Cross-check on the low levels: the values $0, \frac{3}{4}, \frac{3}{2}, 2$ carry multiplicities $1, 2, 2, 2$ (the words $\mathbf{1}; x, y; z, xy; \chi_1(U_1), \chi_1(U_2)$), matching the spectrum used in the weak-coupling tower of §5.

Exact counting function. Substituting $a = 2j_1 + 1$, $b = 2j_2 + 1 \in \mathbb{N}$, the eigenvalue is $(a^2 + b^2 - 2)/4$ with multiplicity $\min(a, b)$, so

$$N(E) = \sum_{a, b \geq 1, a^2 + b^2 \leq 4E + 2} \min(a, b),$$

exact arithmetic with no numerical step.

Asymptotics. With $R^2 = 4E + 2$, comparison of the lattice sum with its integral costs $O(R^2)$ (the summand is bounded by R and varies by at most 1 per unit cell over a domain of $O(R^2)$ cells), and in polar coordinates

$$\iint_{a, b > 0, a^2 + b^2 \leq R^2} \min(a, b) da db = \frac{R^3}{3} \int_0^{\pi/2} \min(\cos \varphi, \sin \varphi) d\varphi = \frac{R^3}{3} (2 - \sqrt{2}),$$

since $\int_0^{\pi/2} \min(\cos, \sin) = 2 \int_0^{\pi/4} \sin = 2 - \sqrt{2}$. With $R^3 = 8E^{3/2}(1 + O(1/E))$,

$$N(E) = \frac{16 - 8\sqrt{2}}{3} E^{3/2} + O(E).$$

The limit exists, so Dixmier measurability holds for this instance and the spectral volume is admissible as an emission (§3.4).

The identity. A three-dimensional geometry of volume V has Weyl coefficient $V\omega_3/(2\pi)^3 = V/(6\pi^2)$, so the volume read from the count is $6\pi^2 \cdot (16 - 8\sqrt{2})/3 = 2\pi^2(16 - 8\sqrt{2}) = (32 - 16\sqrt{2})\pi^2$: exactly the closed form of Appendix A.2. The $\sqrt{2}$ arises on the metric side from the corner integral of Appendix A.2 and on the spectral side from $\int_0^{\pi/2} \min(\cos, \sin) d\varphi$: two unrelated computations forced to one number.

Coupling independence. For $h = \text{Cas} - \lambda M$ with $\|M\| \leq 6$, the operator inequalities $\text{Cas} - 6\lambda \leq h \leq \text{Cas} + 6\lambda$ give $N_{\text{Cas}}(E - c) \leq N_{h - E_0}(E) \leq N_{\text{Cas}}(E + c)$ with $c = 6\lambda + |E_0|$; bounded shifts leave the $E^{3/2}$ coefficient unchanged, so Theorem 4.2 holds at every declared coupling.

Remark A.1 (One-loop warm-up; finite- E diagnostic). The single-plaquette instance closes the same way in dimension one: eigenvalues $j(j+1)$ with multiplicity one give $N(E) = \#\{k \geq 1 : k^2 \leq 4E+1\} \sim 2\sqrt{E}$, and the one-dimensional Weyl law $N \sim (L/\pi)\sqrt{E}$ reads off $L = 2\pi$, the length of $[0, \pi]$ in the metric $4d\theta^2$, matching the one-loop distance formula of §4. As a diagnostic of the subleading (boundary) term in the two-loop count, the exact sum at $E = 10^6$ gives $N(E)/E^{3/2} = 1.5621140$ against the limit $(16 - 8\sqrt{2})/3 = 1.5620972$, the residual carried by the $O(E)$ term.

A.5 Strong coupling

Near the identity class use geodesic normal coordinates $s_i \in \mathbb{R}^3$ per loop ($\theta_i = |s_i|/2$ from $g_{\theta\theta} = 4$). Then $x \approx 2 - |s_1|^2/4$, $y \approx 2 - |s_2|^2/4$, and, since small rotations compose by adding their vectors at leading order, $z \approx 2 - |s_1 + s_2|^2/4$; hence

$$6 - M \approx \frac{1}{4}(|s_1|^2 + |s_2|^2 + |s_1 + s_2|^2), \quad h + 6\lambda \approx -\Delta_{s_1} - \Delta_{s_2} + \frac{\lambda}{4}(s_1^2 + s_2^2 + (s_1 + s_2)^2),$$

with the flat Laplacians of the Γ -metric. Per component the stiffness matrix is $\frac{\lambda}{4} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with eigenvalues $3\lambda/4$ and $\lambda/4$; normal modes $u = (s_1 + s_2)/\sqrt{2}$, $v = (s_1 - s_2)/\sqrt{2}$, three components

each; the ground energy of $-d^2/ds^2 + ks^2$ is $\sqrt{k}$ (spectrum $(2n+1)\sqrt{k}$: level spacing $2\sqrt{k}$, twice the zero-point energy), so

$$E_1 + 6\lambda = 3\sqrt{3\lambda/4} + 3\sqrt{\lambda/4} = \frac{3}{2}(1 + \sqrt{3})\sqrt{\lambda} + O(1),$$

the $O(1)$ collecting measure (drift) and quartic corrections, computed exactly below. Excitations are priced by the level spacing $\omega = 2\sqrt{k}$, not the zero-point energy $\sqrt{k}$. Single quanta of u or v are vectors under the residual diagonal $SO(3)$ and do not lie in the invariant sector, whose functions are built from $|s_1|^2, |s_2|^2, s_1 \cdot s_2$; the lowest invariant excitation is therefore a soft-mode pair at $2\omega_v = 4\sqrt{\lambda/4} = 2\sqrt{\lambda}$, and the next is the mixed scalar $u \cdot v$ at $\omega_u + \omega_v = (1 + \sqrt{3})\sqrt{\lambda}$. Hence the gap-ratio asymptote $(1 + \sqrt{3})/2$ (a ratio of spacings, insensitive to the pricing), and, since v is odd and v^2 even under loop exchange, the swap-even ordering at strong coupling. The virial theorem gives $\lambda\langle 6 - M \rangle = E_{\text{harm}}/2$, i.e. $\langle M \rangle = 6 - \frac{3}{4}(1 + \sqrt{3})/\sqrt{\lambda} + O(1/\lambda)$.

The constant term. The $O(1)$ term is fixed by an exact conjugation. Per $SU(2)$ factor, with $J(s) = [\sin(|s|/2)/(|s|/2)]^2$ the Jacobian of the exponential chart, the Casimir satisfies

$$C_2 = J^{-1/2} \left(-\Delta_{\text{flat}} - \frac{1}{4} \right) J^{1/2}$$

exactly: $J^{1/2}\chi_j = 2\sin((2j+1)|s|/2)/|s|$ is a flat radial eigenfunction at $((2j+1)/2)^2 = j(j+1) + \frac{1}{4}$ for every j (verified symbolically through $2j = 6$). Hence $h + 6\lambda$ is unitarily a *flat* six-dimensional problem,

$$h + 6\lambda \cong -\Delta_6 - \frac{1}{2} + \lambda W, \quad W = 6 - M \text{ in normal coordinates,}$$

with W exact trigonometry: from $\text{tr}(U_1 U_2) = 2[\cos u_1 \cos u_2 - (\hat{s}_1 \cdot \hat{s}_2) \sin u_1 \sin u_2]$, $u_i = |s_i|/2$,

$$W = \frac{1}{4}(|s_1|^2 + |s_2|^2 + |s_1 + s_2|^2) - \frac{|s_1|^4 + |s_2|^4}{96} - \frac{|s_1|^2 |s_2|^2}{32} - \frac{(s_1 \cdot s_2)(|s_1|^2 + |s_2|^2)}{48} + O(s^6),$$

the quadratic part reproducing the stiffnesses above. In the normal modes $u = (s_1 + s_2)/\sqrt{2}$, $v = (s_1 - s_2)/\sqrt{2}$ the harmonic ground state is the product Gaussian with per-component variances $\sigma_u^2 = 1/\sqrt{3\lambda}$, $\sigma_v^2 = 1/\sqrt{\lambda}$, and first-order perturbation theory in the quartic gives $\lambda\langle W_4 \rangle = -(30 + 13\sqrt{3})/192$; second order in W_4 and first order in W_6 are both $O(\lambda^{-1/2})$. With the exact $-\frac{1}{2}$ from the conjugation,

$$c_0 = -\frac{1}{2} - \frac{30 + 13\sqrt{3}}{192} = -\frac{126 + 13\sqrt{3}}{192} \approx -0.77352.$$

Controls: (i) the one-loop analogue of the identical mechanics predicts $c_0^{(1)} = -\frac{1}{4} - \frac{5}{64} = -\frac{21}{64}$, and the exact character-basis tridiagonal reproduces it to 10^{-9} under a large- λ fit; (ii) every algebraic step (trace identity, quartic expansion, Gaussian moments, conjugation eigenfunctions) verifies symbolically; (iii) converged exact-moment variational values at $\lambda \in \{2, 4, 8\}$ sit within 0.014 of c_0 with subleading coefficients of natural size, and, with c_0 fixed, the implied (c_{-1}, c_{-2}) postdict the certified $\lambda = 1/2$ window; (iv) the derived value lies inside the certified $\lambda = 2$ window outright.

A.6 The fourth-order weak-coupling coefficient

At $\lambda = 0$ the ground state is the Haar state with $E_0 = 0$ and $\langle M \rangle = 0$. The span of invariant monomials of degree ≤ 2 (ten elements) is invariant under the Casimir, closing by the Γ -calculus of §5, and decomposes exactly with eigenvalues $0, \frac{3}{4}, \frac{3}{2}, 2, \frac{11}{4}, 4$ and multiplicities $1, 2, 2, 2, 2, 1$, matching the combinatorial audit $\min(2j_1, 2j_2) + 1$ of Appendix A.4. Every intermediate state of fourth-order Rayleigh–Schrödinger perturbation theory lies in this span (each application of M raises monomial degree by at most one and the path must return to the vacuum), so the fourth order is a finite computation in exact rational arithmetic with the reduced resolvent $R = -\sum_{\mu \neq 0} P_\mu / \mu$:

$$E^{(4)} = \langle 0 | V R V R V R V | 0 \rangle - E^{(2)} \langle 0 | V R^2 V | 0 \rangle = -\frac{3409}{297} - \left(-\frac{10}{3}\right)(4) = \frac{551}{297},$$

with $V = -M$ per unit coupling and $\langle 0 | V R^2 V | 0 \rangle = 4$ exactly. The same machinery reproduces $E^{(2)} = -\frac{10}{3}$ and $E^{(3)} = -\frac{32}{9}$, and an independent variational route (exact-moment Rayleigh–Ritz at monomial degree ≤ 6 , couplings $\lambda \in [1/96, 1/32]$) confirms the value.

The fifth order follows from the same objects by the $2n+1$ rule: with $\psi_1 = R V | 0 \rangle$ and $\psi_2 = R V R V | 0 \rangle$ (both inside the degree- ≤ 2 span, so no further resolvent is needed),

$$E^{(5)} = \langle \psi_2 | V | \psi_2 \rangle - 2E^{(2)} \langle \psi_1 | \psi_2 \rangle - E^{(3)} \langle \psi_1 | \psi_1 \rangle = \frac{172064}{9801} \approx +17.5558,$$

confirmed variationally: with c_4 fixed, the residual fit gives $c_5 = 17.579$, and the diagnostic ratio resid/λ^5 marches monotonically onto the exact value as $\lambda \rightarrow 0$.

A caution on reading coefficients off bracket data: the higher-order coefficients here are large, so naive fits mislead—a two-point fit at $\lambda \approx 1/16$ omitting the fifth order reports $c_4 + c_5 \lambda \approx 3$ rather than $c_4 \approx 1.86$, and a joint fit of (c_4, c_5) leaks $c_6 \approx 21$ into the fifth-order estimate. Certified brackets bound energies, not series coefficients; the coefficients must be derived, with the brackets as their check.

B The first post-Newtonian two-body read

The velocity structure derived below was first obtained from one-graviton exchange by Iwasaki [19]; the derivation is retained because the sector as declared (forced source, no dynamical self-stress vertex, Definition 11.16) must be shown to produce it, and because the abelian control pins the conventions. Between conserved sources the single-exchange kernel is unique at this order (gauge-dependent parts of the propagator drop against conservation); the propagator algebra below is evaluation scaffolding for a chart-free read, the two-body interaction energy.

Instantaneous part. Worldline stress $T_i^{\mu\nu} = m_i u_i^\mu u_i^\nu / \gamma_i$ (the $1/\gamma$ from the worldline density); spin-2 numerator $P_{\mu\nu\rho\sigma} = \frac{1}{2}(\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho} - \eta_{\mu\nu}\eta_{\rho\sigma})$, so $T_1 P T_2 = \frac{1}{2} m_1 m_2 [2(u_1 \cdot u_2)^2 - 1] / (\gamma_1 \gamma_2)$. Normalizing the static term to $G m_1 m_2 / r$ and expanding $u_1 \cdot u_2 = \gamma_1 \gamma_2 (1 - v_1 \cdot v_2)$:

$$\begin{aligned} \frac{2(u_1 \cdot u_2)^2 - 1}{\gamma_1 \gamma_2} &= 1 + \frac{3}{2}(v_1^2 + v_2^2) - 4 v_1 \cdot v_2 + O(v^4), \\ L_{\text{inst}} &= \frac{G m_1 m_2}{r} + \frac{G m_1 m_2}{2r} [3(v_1^2 + v_2^2) - 8 v_1 \cdot v_2]. \end{aligned}$$

Retardation. $1/(k^2 - \omega^2) = 1/k^2 + \omega^2/k^4 + O(\omega^4)$, with $1/k^2 \leftrightarrow 1/(4\pi r)$ and $1/k^4 \leftrightarrow -r/(8\pi)$. The ω^2 term between the static channels is $\Delta L = -\frac{G m_1 m_2}{2} \partial_t \partial_{t'} r(t, t')|_{t'=t}$ with

$r(t, t') = |x_1(t) - x_2(t')|$; since $\partial_t \partial_{t'} r = -[v_1 \cdot v_2 - (n \cdot v_1)(n \cdot v_2)]/r$,

$$\Delta L = +\frac{Gm_1 m_2}{2r} [v_1 \cdot v_2 - (n \cdot v_1)(n \cdot v_2)].$$

Sum.

$$L_{\text{inst}} + \Delta L = \frac{Gm_1 m_2}{r} + \frac{Gm_1 m_2}{2r} [3(v_1^2 + v_2^2) - 7v_1 \cdot v_2 - (n \cdot v_1)(n \cdot v_2)],$$

the Einstein–Infeld–Hoffmann velocity structure [65].

Abelian control. The same two lines in the $U(1)$ sector: current numerator $(u_1 \cdot u_2)/(\gamma_1 \gamma_2) = 1 - v_1 \cdot v_2$ exactly, static sign reversed, retardation kernel identical. Then $L = -q_1 q_2/r + (q_1 q_2/r) v_1 \cdot v_2 - (q_1 q_2/2r)[v_1 \cdot v_2 - (n \cdot v_1)(n \cdot v_2)] = -q_1 q_2/r + (q_1 q_2/2r)[v_1 \cdot v_2 + (n \cdot v_1)(n \cdot v_2)]$: the Darwin Lagrangian [67], a known result, validating the kernel identity and the sign conventions.

C Executed falsifier battery

Every check computes a ground truth by an independent route (exact diagonalization, matrix logarithm, or Fock-space projection) and compares it to the closed form; no empirical input enters. The closures instantiate Theorems 2.10–2.11: where a bracket closes to $L^* = U^*$ the certificate factors annihilate the state, and that annihilation is verified.

Free-field zero. The level-1 optimizer of $\text{Tr}(h\gamma)$ is the negative-energy projector (the Dirac sea as output) to 0; the subtracted vacuum energy is 0; the charge-graded antiparticle energies are strictly positive (vacuum stable), catching any grading-sign error.

van Hove. The completed square of Definition 2.3 is an operator identity to 3.6×10^{-14} , positive-semidefinite, and the exact-diagonalization ground energy meets the certificate value $-\sum |B|^2/\omega$ to 0: the boson word calculus verified, and tight ($L^* = U^*$).

Modular reads. For a two-mode Gaussian state built by a Gaussian unitary on a thermal state (no truncation assumption), the modular Hamiltonian extracted from $-\log \rho$ is quadratic (evaluability), its symplectic-log relation $\nu = \frac{1}{2} \coth(g/2)$ holds, and the state is KMS at inverse modular temperature one; all three converge with Fock cutoff ($2 \times 10^{-3} \rightarrow 6 \times 10^{-5}$), confirming physics rather than artifact. The rank-deficiency fail-shut and the two-mode Gaussian modular flow (KMS to machine precision) are the temporal end-to-end pins.

Two-mode Gaussian conditional (exact, no truncation). (a) The total-charge conditional expectation commutes with the per-covolume average to 0.0; a symmetry-breaking per-mode conditioning fails to commute at 0.41, so the commutation test discriminates within the conditioning class admitted by Definition 11.6. (b) The charge-conditioned two-point function from the natural-orbital closed form equals the Fock ground truth to 1.4×10^{-16} , and the certified two-sided bracket closes on the conditional ground energy, $L^* = U^* = e_0$, certificate gap 0: boundary contact (Theorem 2.11) exhibited.

Emergence (harmonic half-pin). Zero-point energy exact by two routes; the structure-factor read develops Bragg atoms at the reciprocal-lattice vectors with the correct Debye–Waller intensity ratio. Half-pin: the lattice is in the coefficients, so this tests the read, not order-from-disorder, which needs the interacting engine.

Cost-Theorem solvable models. RPA per-mode saturation (the variational bound saturates the exact per-mode gap; totals $\sim r_0^{-3}$); two-mode order structure (first-order vanishing); three-mode commutator fluctuation ($O(\xi^2 s^2)$); four-mode conditioning law ($A = 2\lambda^2/\omega_2$, flat

reduced densities have zero f-sum so conditioning is load bearing); screened incoherent weight ($\sim r_0^{-6}$); the harmonic crystal and non-adiabatic H₂/HD massive pins; van Hove, the harmonic charge, and spin–boson/Rabi radiative pins.

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